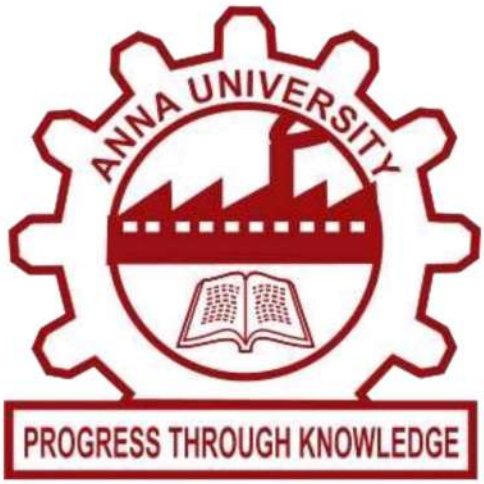
UNIVERSITY COLLEGE OF ENGINEERING NAGERCOIL

(ANNA UNIVERSITY CONSTITUENT COLLEGE)

KONAM, NAGERCOIL - 629004



RECORD NOTE BOOK

EMBEDDED SYSTEMS

AND IOT - CS3691

Register No :

Name :

Year/Semester :

Department :

UNIVERSITY COLLEGE OF ENGINEERING NAGERCOIL

(ANNA UNIVERSITY CONSTITUENT COLLEGE)

KONAM, NAGERCOIL -629004

Register No:

Certified that, this is the bonafide record of work done by Mr. / Ms. ..... Of VI Semester in Computer Science and Engineering of this college, in the Embedded Systems and Iot (CS3691) during academic year (2025-2026) in partial fulfilment of the requirements of the B.E Degree course of the Anna University Chennai.

Staff-in-charge **Head of the Department**

This record is submitted for the University Practical Examination held on .....

Internal Examiner **External Examiner**

LIST OF EXPERIMENTS

S.No	Date	Name of the Experiments	Page Number	Sign

S.No	Date	Name of the Experiments	Page Number	Sign

EX NO : DATE :	**WRITE 8051 ASSEMBLY LANGUAGE EXPERIMENTS USING KEIL SIMULATOR**

Aim:

To write 8051 assembly language experiments using simulator.

Apparatus required:

Keil µ software

Procedure:

1. The 8051 microcontroller has a set of instructions for arithmetic, logic, and data manipulation operations. that demonstrates some basic ALU operations using assembly language for the 8051 microcontroller.

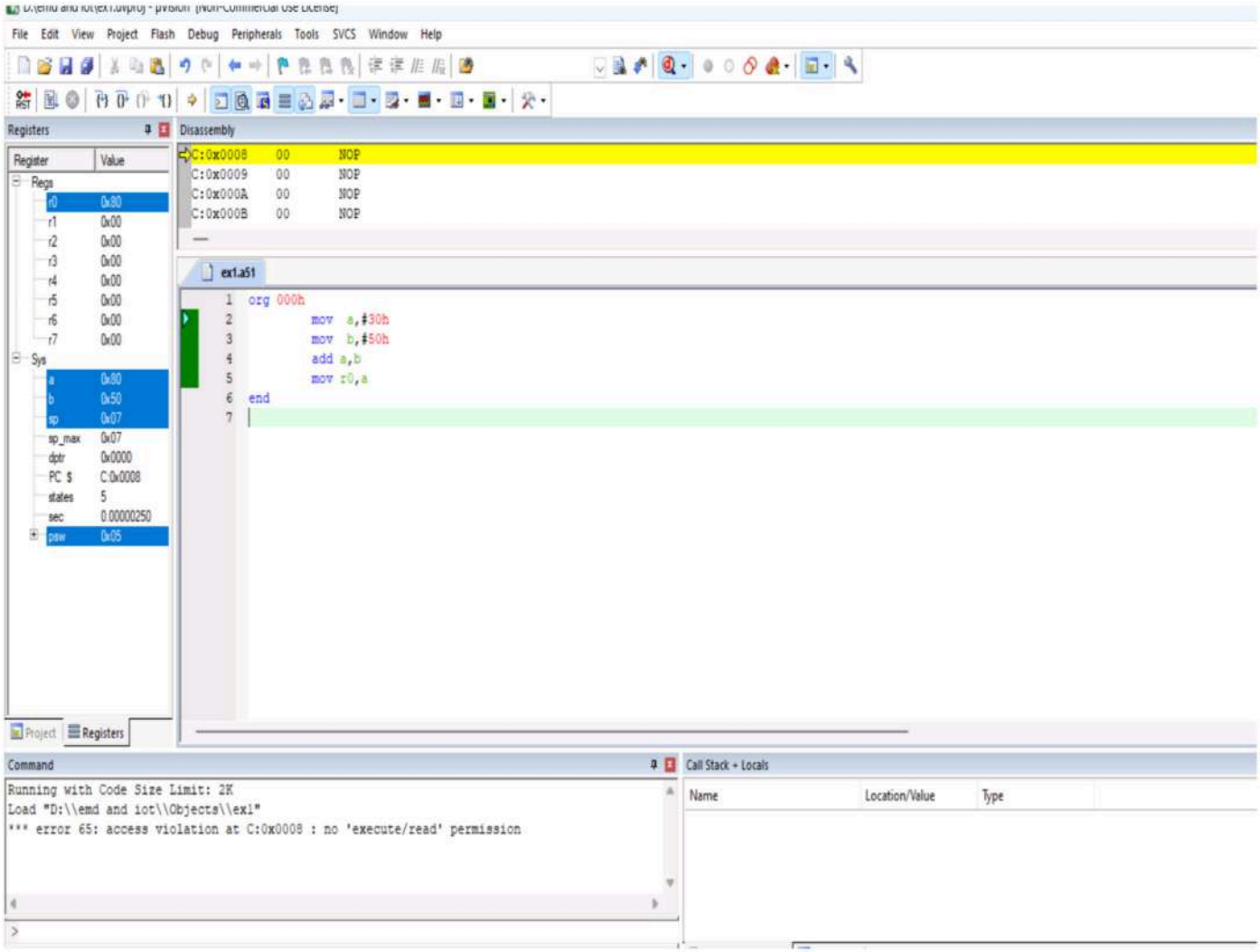
2. The various ALU operations, including addition, subtraction, multiplication, division, logical AND/OR/XOR, and rotation. Note that the specific instructions and registers used may vary depending on the exact 8051 microcontroller variant you are working with.

3. Always refer to the datasheet or reference manual for your specific microcontroller to ensure accurate programming.

Program:

```
org 000h

mov  a,#30h

mov  b,#50h

add a,b

mov r0,a

end
```

Output:



Result:

Thus the 8051 assembly language program was successfully implemented and simulated.

EX NO : DATE :	**WRITE 8051 ASSEMBLY LANGUAGE EXPIREMENTS USING EDSIM 51 SIMULATOR**

Aim:

The purpose of this experiment is to write and implement 8051 assembly language experiments using simulator.

Apparatus required:

Edsim51 simulator

Program logic:

An Assembly language program consists of, among other things, a series of line of Assemble language instructions. An Assembly language instruction consists of mnemonic, optionally followed by one or two operands. The operands are the data items being manipulated and the mnemonics are the commands to the CPU, telling it what to do with those items.

□ Looping:

Repeating a sequence of instructions a certain number of time is called a loop. The loop is one of the most widely used actions that any microprocessor performs. In this instruction, the register is decremented; if it is not zero, it jumps to the target address referred by the label. Prior to the start of the loop the register is loaded with the counter for the number of repetitions. In this instruction both the register decrement and the decision to jump are combined into a single instruction.

A loop inside another loop is called nested loop.

□ Unconditional jump instruction:

The unconditional jump is a jump in which control is transferred unconditionally to the target location. In 8051 there is two unconditional jumps:

- **LJMP(long jump):**

LJMP is an unconditional long jump. It is a 3-byte instruction in which the first byte is a opcode, second and third bytes represent the 16-bit address of the target location. The 2-byte target address allows a

jump to any memory location from 0000 to FFFFH.

- **SJMP(Short jump):**

SJMP is an unconditional short jump. It is a 2-byte instruction in which the first byte is a opcode and second bye is the relative address of the target location. The target address ranges from 00 to FFH.

Procedure:

Step-1: Open the Edsim51 simulator.

Step-2: Type the program and save it with ".asm" or ".hex" extension.

Step-3: Run the program and rectify the result.

Step-4: Observe the output.

Step-5: Repeat the above steps for all the program.

Program:

1. Sample of Assembly language program:

```
ORG 0H
MOV R5, #25H
MOV R7,#34H
MOV A, #0
ADD A, R5
ADD A, R7
ADD A, #12H
HERE:SJMP HERE
END
```

2. Multiply 25 by 10 using the technique of repeated addition:

```
MOV A, #0
MOV R2, #10
AGAIN:ADD A, #25
DJNZ R2, AGAIN
MOV R5,A
```

3. **Program to add first 10 natural number:**

```
MOV A,#0
MOV R2, #10
MOV R0, #0
AGAIN:INC R0
ADD A,R0
DJNZ R2, AGAIN
MOV 46H, A
```

4. **Program to load the accumulator with the value of 55H and complement the ACC 700 times:**

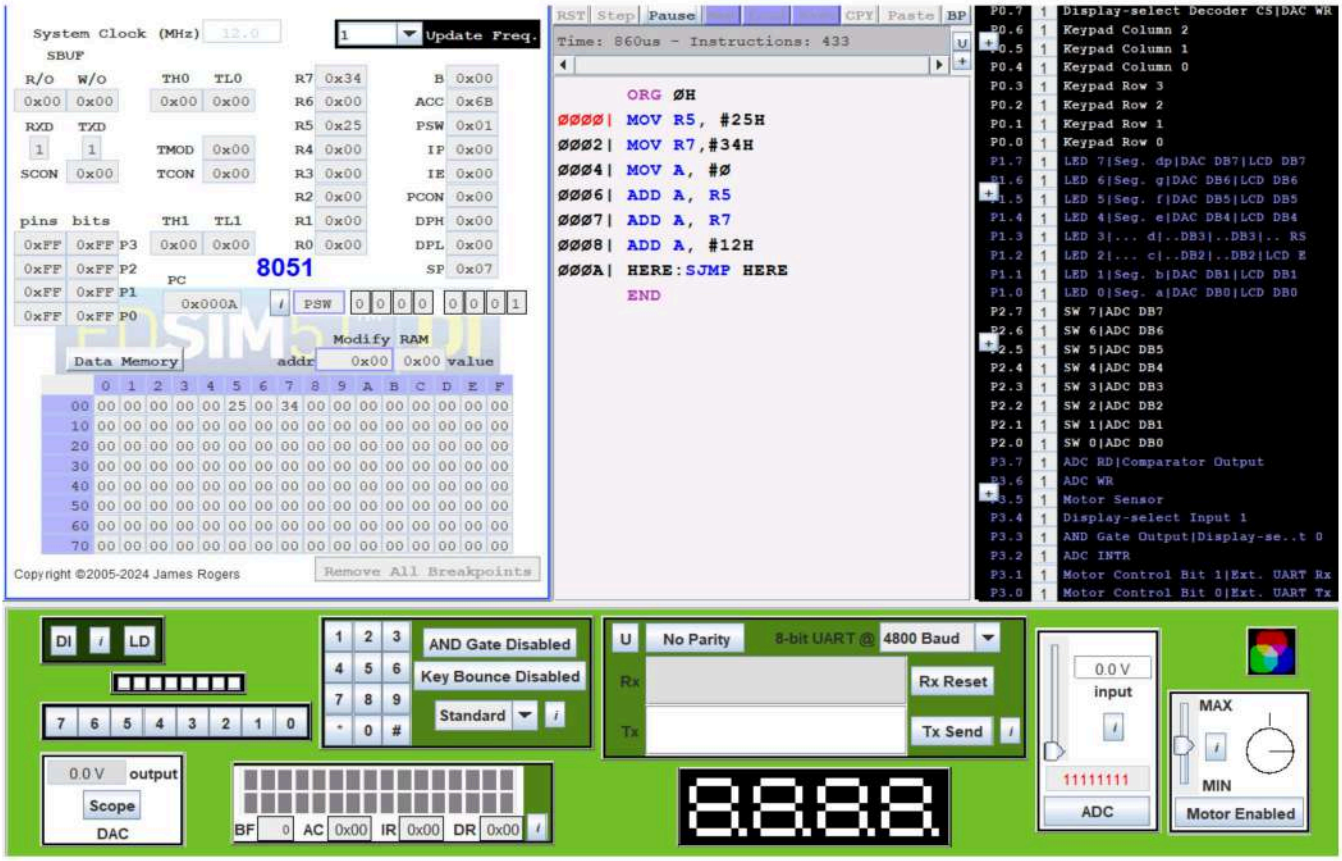
```
MOV A,#55H
MOV R3,#10
NEXT:MOV R2,#70
AGAIN:CPL A
DJNZ R2,AGAIN
DJNZ R3,NEXT
```

5. **Multiply ECH by 25H using the technique of repeated addition:**

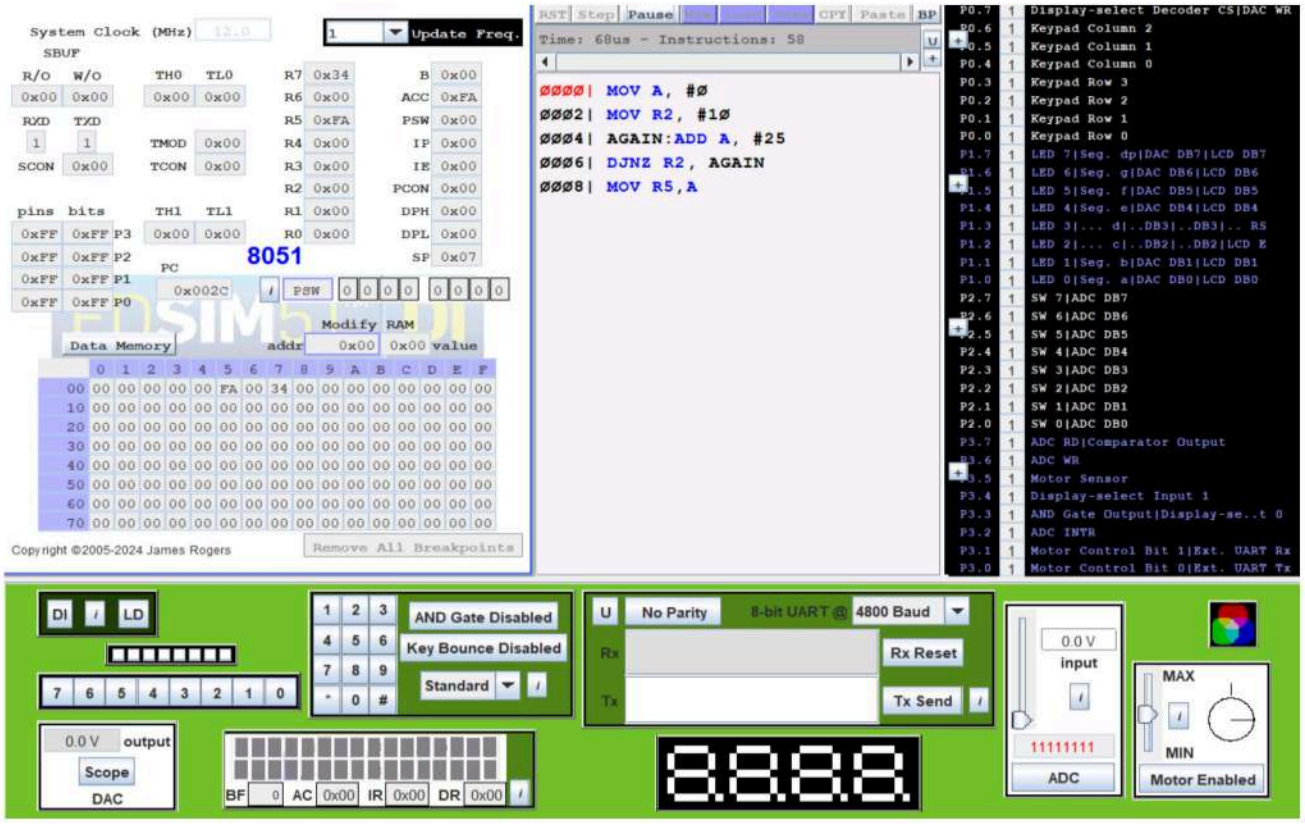
```
MOV R1,#0
MOV A,#0
MOV R0,#25H
AGAIN: ADD A,#0ECH
JNC HERE
INC R1
HERE: DJNZ R0,AGAIN
MOV R0,A
```

Output:

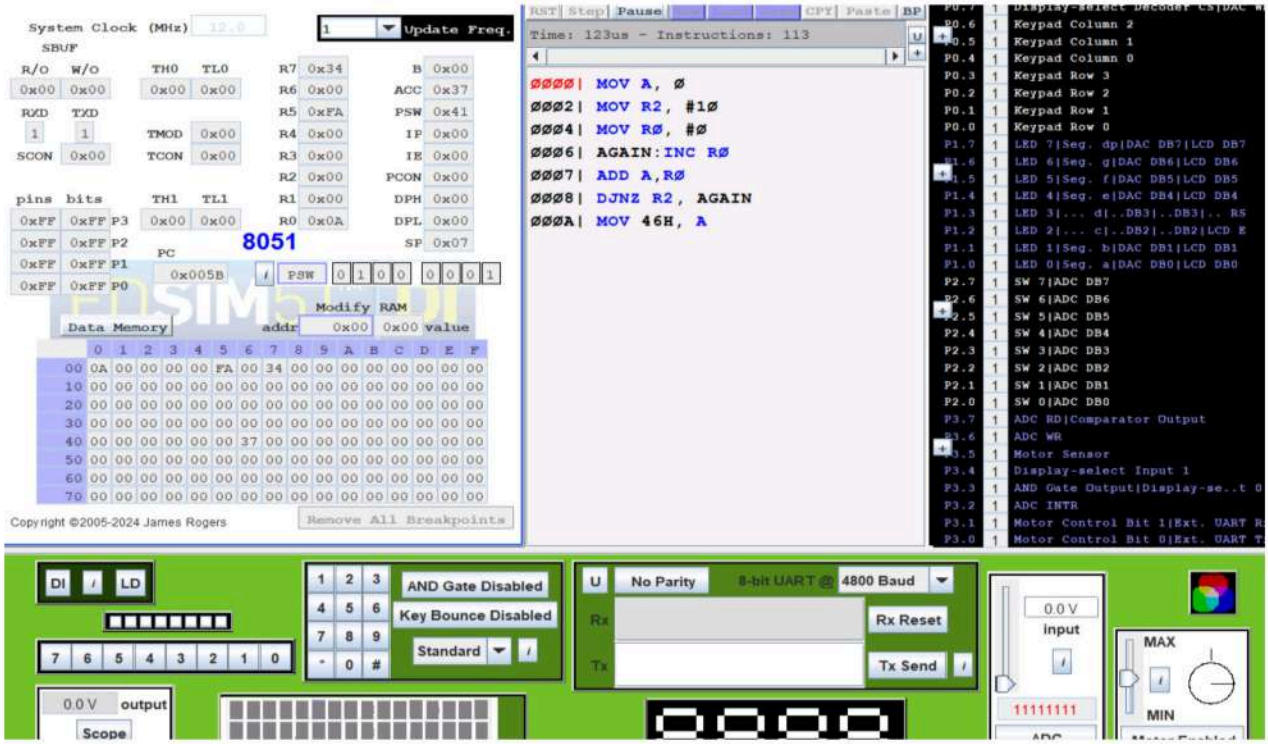
1. Sample of Assembly language program:



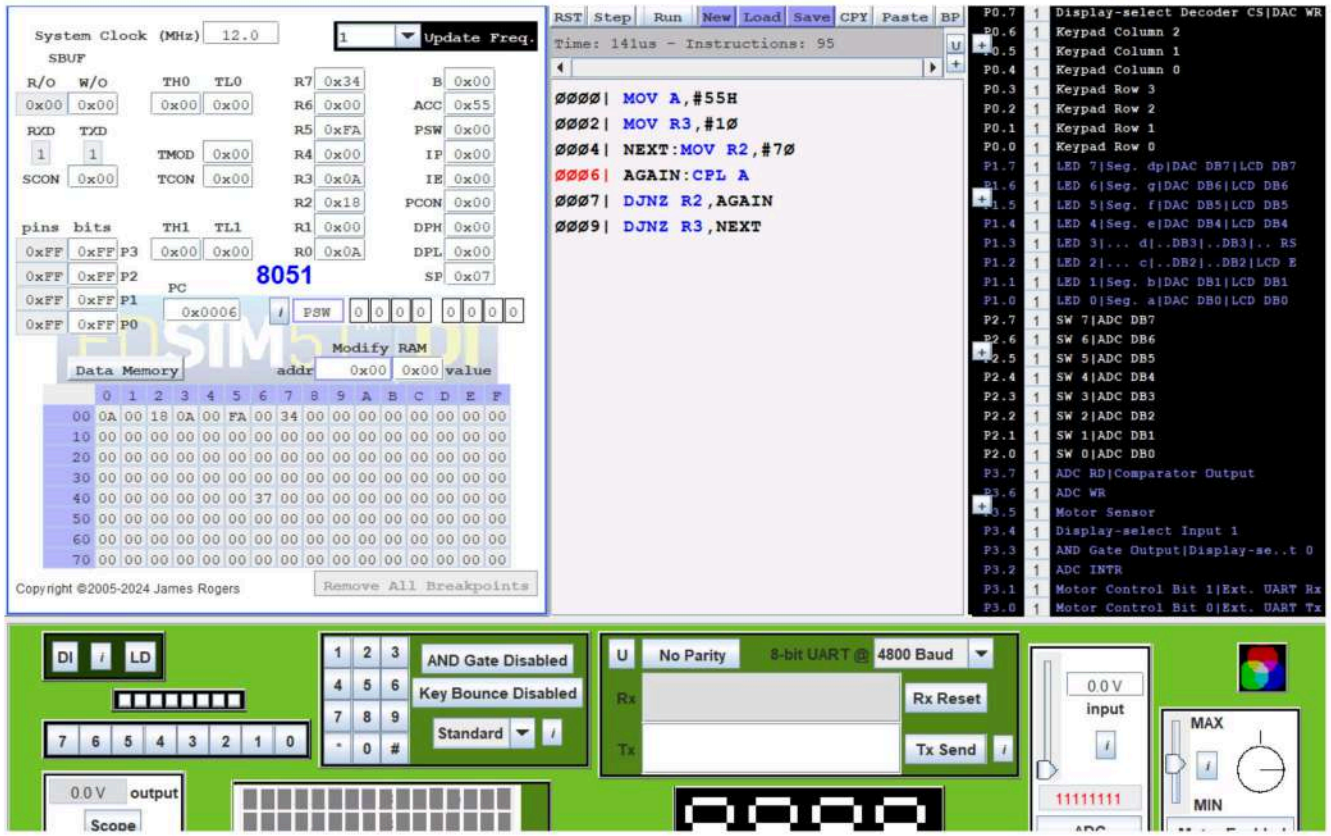
2. Multiply 25 by 10 using the technique of repeated addition:



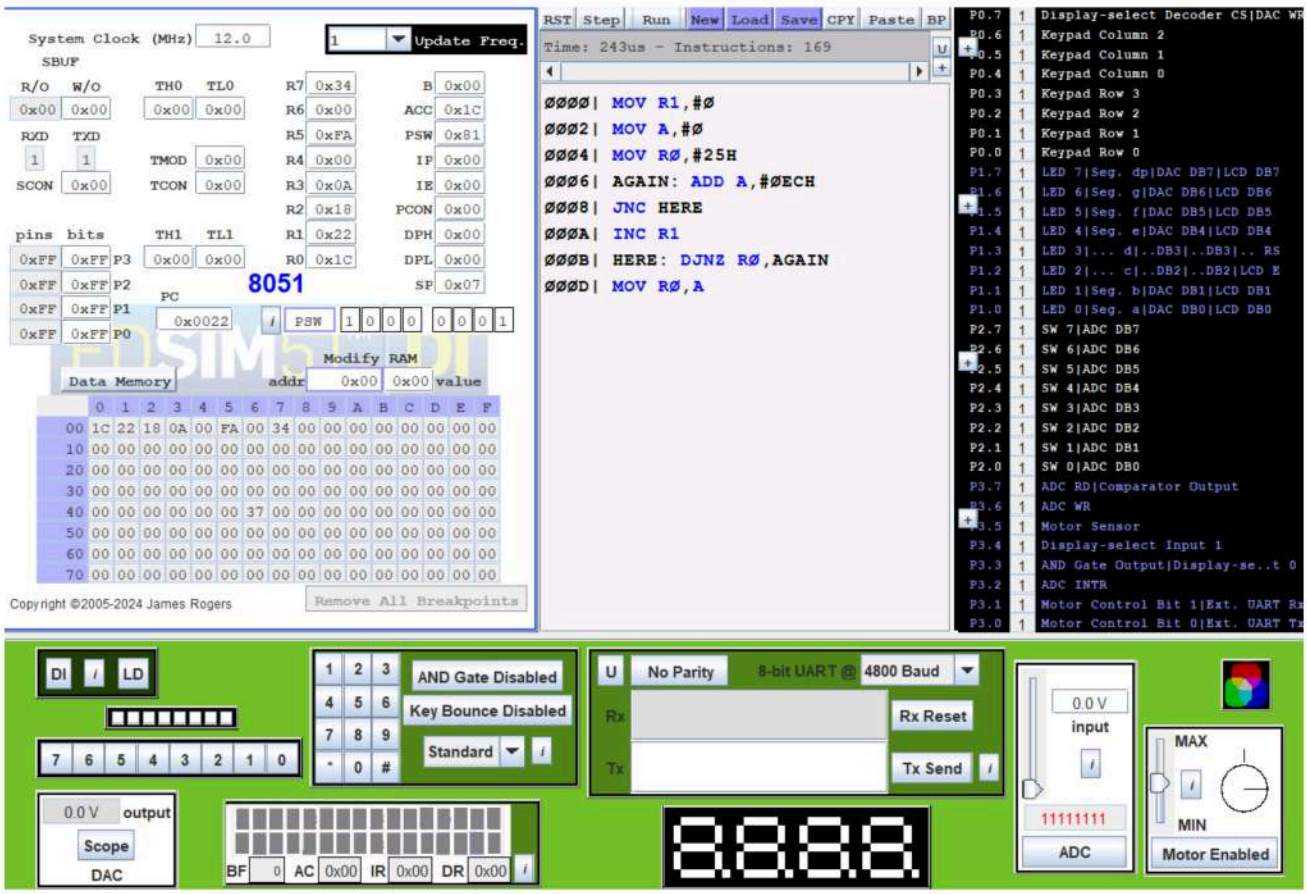
3. Program to add first 10 natural number:



4. Program to load the accumulator with the value of 55H and complement the ACC 700 times:



5. Multiply ECH by 25H using the technique of repeated addition:



Result:

Thus the 8051 assembly level program was written and implemented successfully

EX NO : DATE :	**TEST DATA TRANSFER BETWEEN REGISTER AND MEMORY USING KEIL SIMULATOR**

Aim:

To test data transfer between register and memory using keil simulator

Apparatus required:

Keil µ software

Procedure:

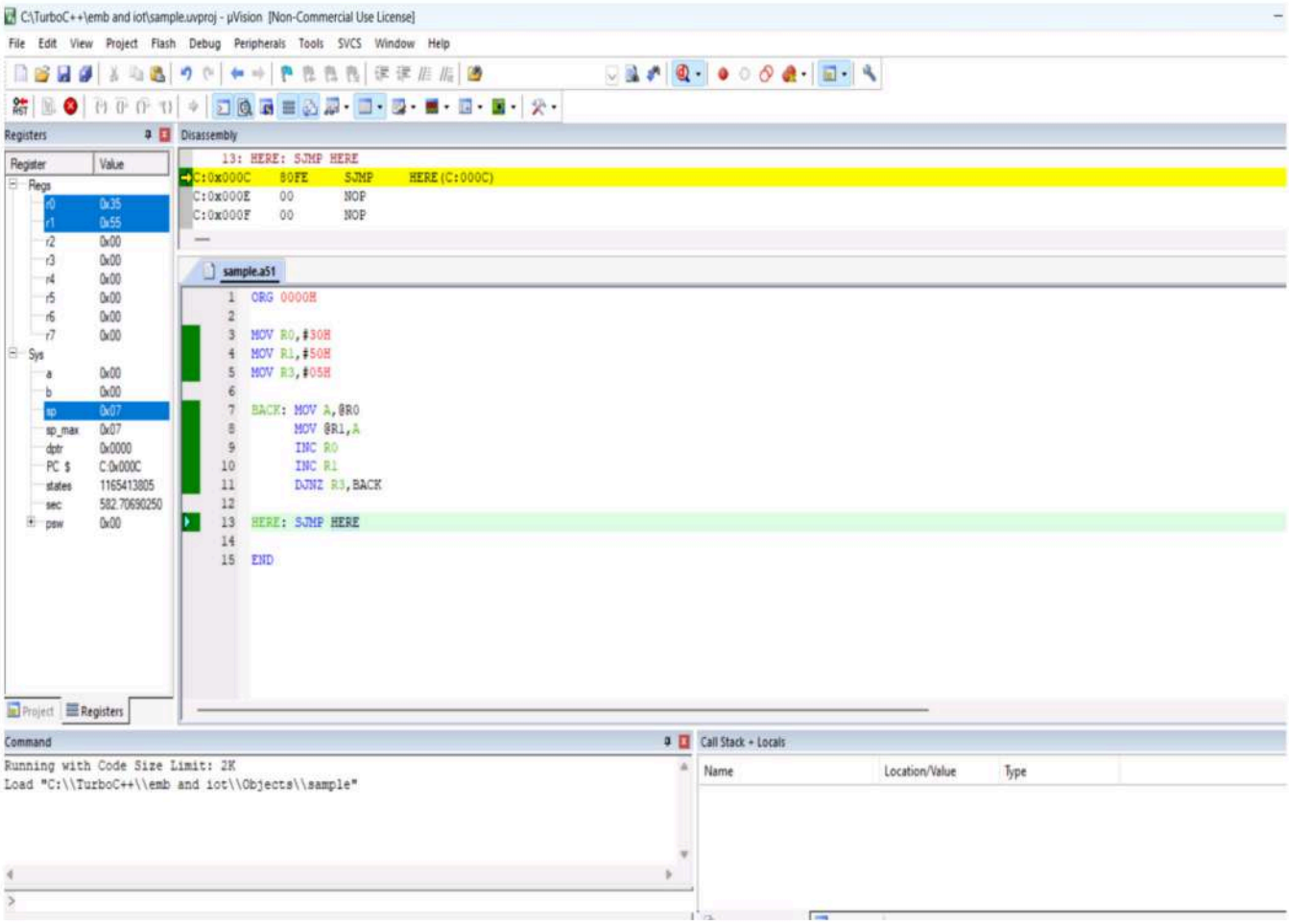
1. The 8051 microcontroller has a set of instructions for arithmetic, logic, and data manipulation operations. that demonstrates some basic ALU operations using assembly language for the 8051 microcontroller.
2. The various ALU operations, including addition, subtraction, multiplication, division, logical AND/OR/XOR, and rotation. Note that the specific instructions and registers used may vary depending on the exact 8051 microcontroller variant you are working with.
3. Always refer to the datasheet or reference manual for your specific microcontroller to ensure accurate programming.

Program:

```
ORG 0000H

MOV R0,#30H

MOV R1,#50H

MOV R3,#05H

BACK: MOV A,@R0

MOV @R1,A

INC R0

INC R1

DJNZ R3,BACK

HERE: SJMP HERE
```

END

Output:



Result:

The program successfully performed simple data transfer from one memory location to another using registers.

EX NO : DATE :	TO TEST DATA TRANSFER BETWEEN REGISTER AND MEMORY USING EDSIM 51 SIMULATOR

Aim:

The purpose of this experiment is to text data transfer between register and memory.

Software required:

Edsim51 simulator.

Program logic:

In 8051, the action of comparing and jumping are combined into single instruction called CJNE. The CJNE instruction compares two operands and jump if they are not equal. In addition, it changes the CY flag to indicate if the destination operand is larger or smaller. It is important to notice that the operand themselves remains unchanged.

Procedure:

Step-1: Enter the opcode in Edsim51 simulator.

Step-2: Execute the program.

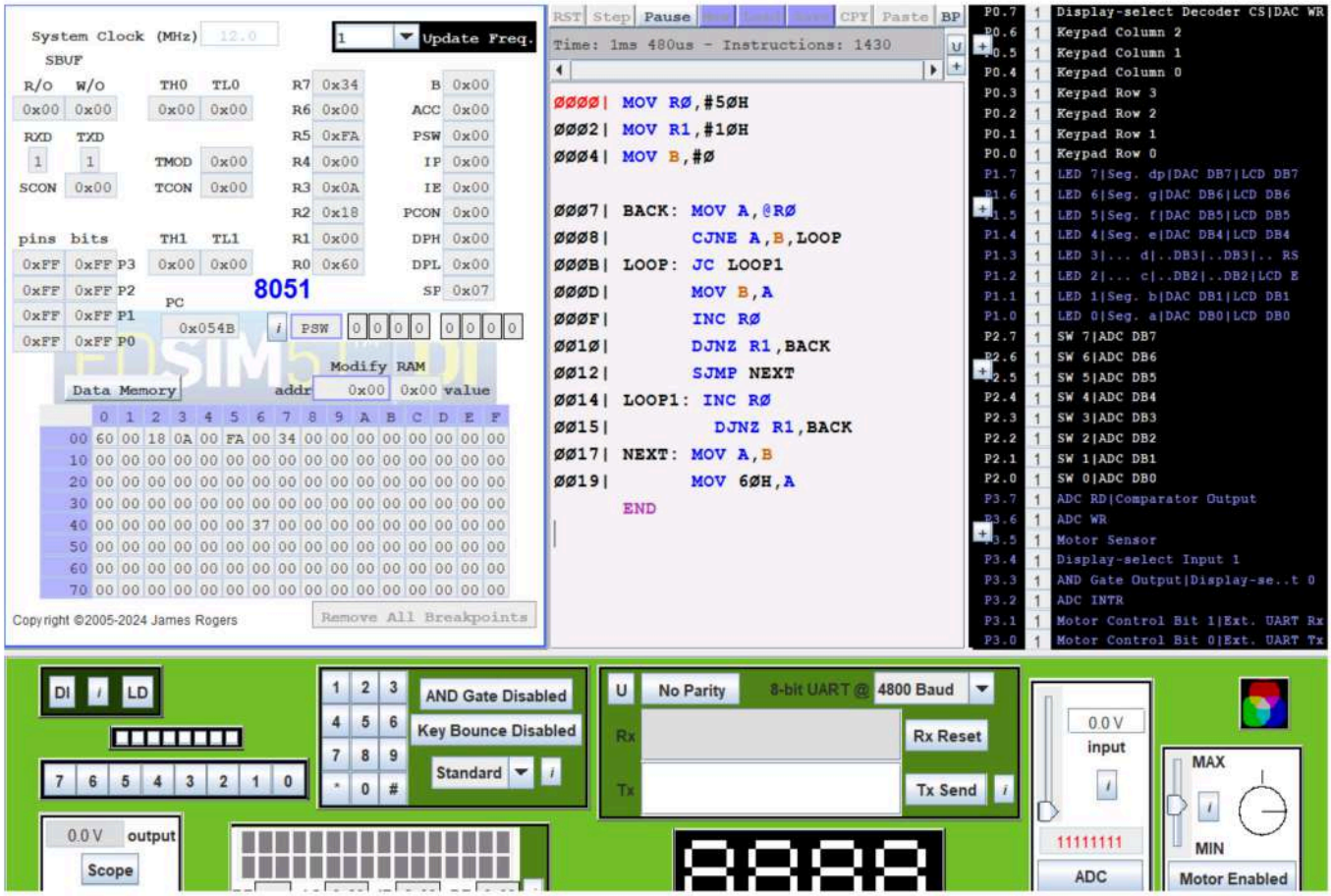
Step-3: Check the result in register A.

Program:

```
MOV R0,#50H
MOV R1,#10H
MOV B,#0
BACK: MOV A,@R0
      CJNE A,B,LOOP
LOOP: JC LOOP1
      MOV B,A
      INC R0
```

```
        DJNZ R1,BACK
        SJMP NEXT
LOOP1: INC R0
        DJNZ R1,BACK
NEXT: MOV A,B
        MOV 60H,A
END
```

Output:



Result:

Thus the program to text data transfer between Register and Memory was written and executed successfully.

EX NO: DATE:	**PERFORM ALU OPERATIONS**

Aim:

To perform ALU operations using Keil software.

Apparatus required:

Keil µ software

Procedure:

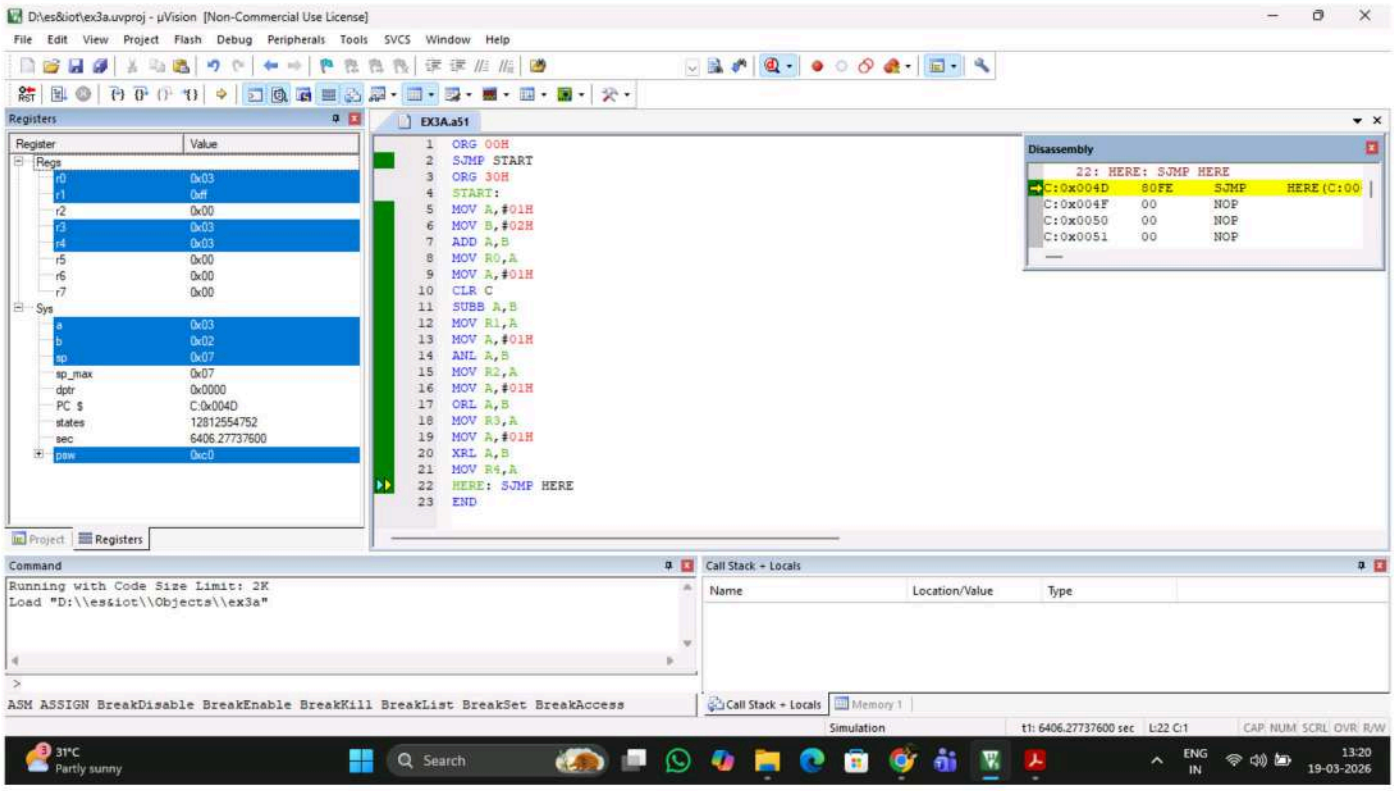
1. The 8051 microcontroller has a set of instructions for arithmetic, logic, and data manipulation operations that demonstrates some basic ALU operations using assembly language for the 8051 microcontroller.
2. The various ALU operations, including addition, subtraction, multiplication, division, logical AND/OR/XOR, and rotation. Note that the specific instructions and registers used may vary depending on the exact 8051 microcontroller variant you are working with.
3. Always refer to the datasheet or reference manual for your specific microcontroller to ensure accurate programming.

Program:

```
ORG 00H
SJMP START
ORG 30H
START:
MOV A,#01H
MOV B,#02H
ADD A,B
MOV R0,A
MOV A,#01H
CLR C
```

```
SUBB A,B
MOV R1,A
MOV A,#01H
ANL A,B
MOV R2,A
MOV A,#01H
ORL A,B
MOV R3,A
MOV A,#01H
XRL A,B
MOV R4,A
HERE: SJMP HERE
END
```

Output:



Result:

Thus the ALU operations using Keil software was performed and the result was successfully simulated.

EX NO: DATE :	**PERFORM ALU OPERATION USING 8051 MICROCONTROLLER**

Aim:

The purpose of this experiment is to Add, Subtract, Multiply and Divide the given two 8 bit numbers and store them in a memory location using Edsim 51 Simulator.

Software requirement:

Edsim 51 Simulator

Procedure:

Step-1: Enter the opcodes from memory location 4200.

Step-2: Executes the program.

Step-3: Check for the result at 4100 and 4101. Using the accumulator, Subtraction is performed and the result is stored. Immediate addressing is employed. The SUBB instruction drives the result in the accumulator.

Program:

ADDITION:

```
CLR C           ; CY = 0
MOV A, #45H     ; Low byte 1
ADD A, #0ECH    ; Add low byte 2
MOV R0, A       ; Store low byte result
MOV A, #02H     ; High byte 1
ADDC A, #0FCH   ; Add high byte 2 WITH carry
MOV R1, A       ; Store high byte result
```

SUBTRACTION:

Subtraction with CY=0:

```
ORG 00H
MOV A, #50H        ; Load first number (minuend)
CLR C              ; Ensure CY = 0 (important!)
SUBB A, #20H       ; A = 50H - 20H
MOV R0, A          ; Store result
END
```

Subtraction with CY=1:

```
ORG 00H
MOV A, #50H        ; A = 50H
SETB C             ; CY = 1 (important!)
SUBB A, #20H       ; A = 50H - 20H - 1
MOV R0, A          ; store result
END
```

MULTIPLICATION:

```
ORG 00H
MOV A, #05H        ; A = 5
MOV B, #04H        ; B = 4
MUL AB             ; A × B
MOV R0, A          ; store lower byte
MOV R1, B          ; store higher byte
END
```

DIVISION:

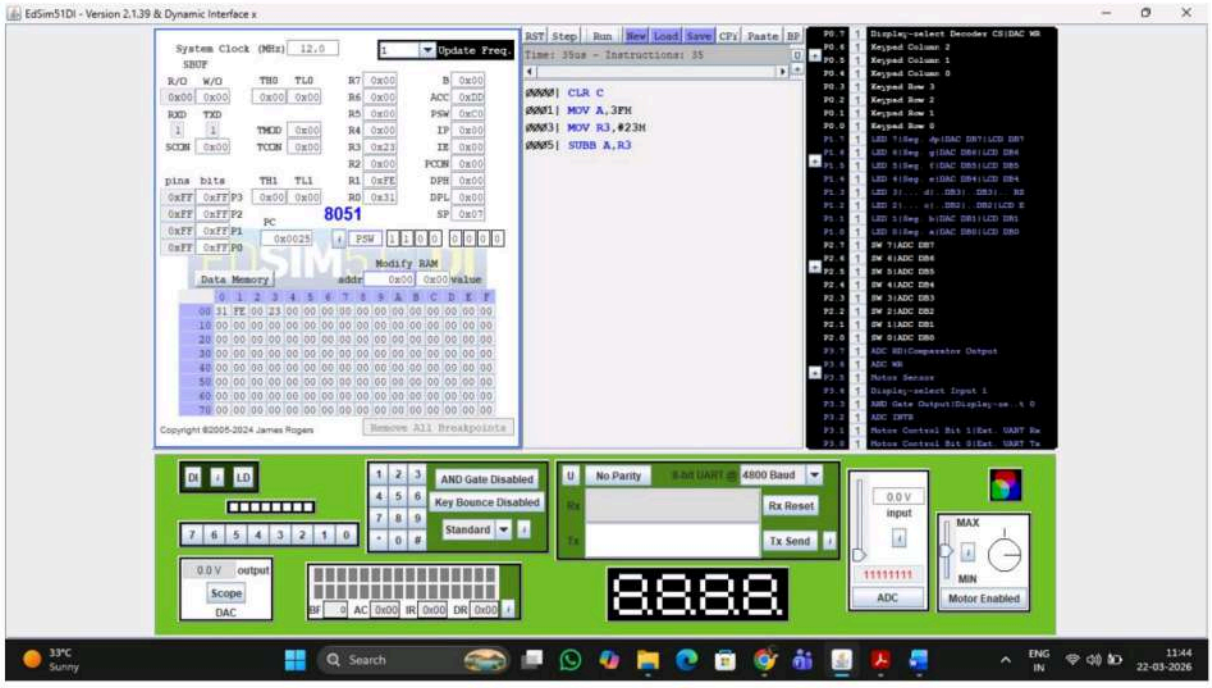
```
ORG 00H
MOV A, #14H        ; A = 20 (decimal)
MOV B, #05H        ; B = 5
DIV AB             ; A / B
MOV R0, A          ; Quotient
MOV R1, B          ; Remainder
END
```

Output:

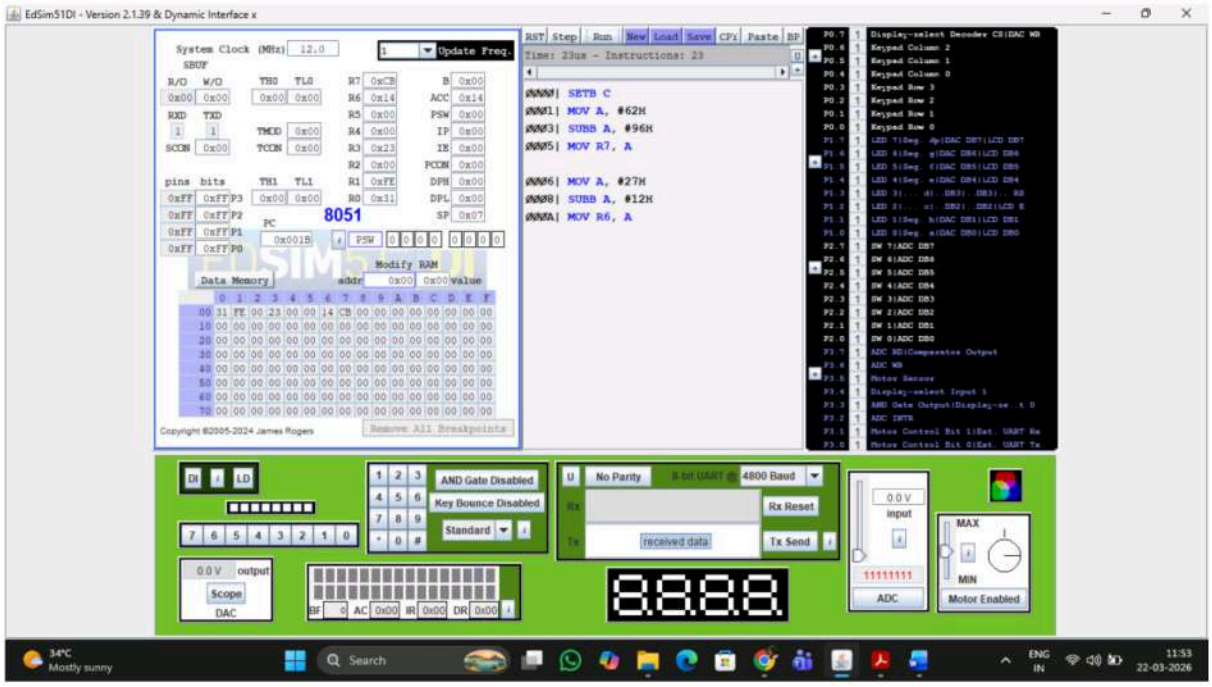
ADDITION:

SUBTRACTION:

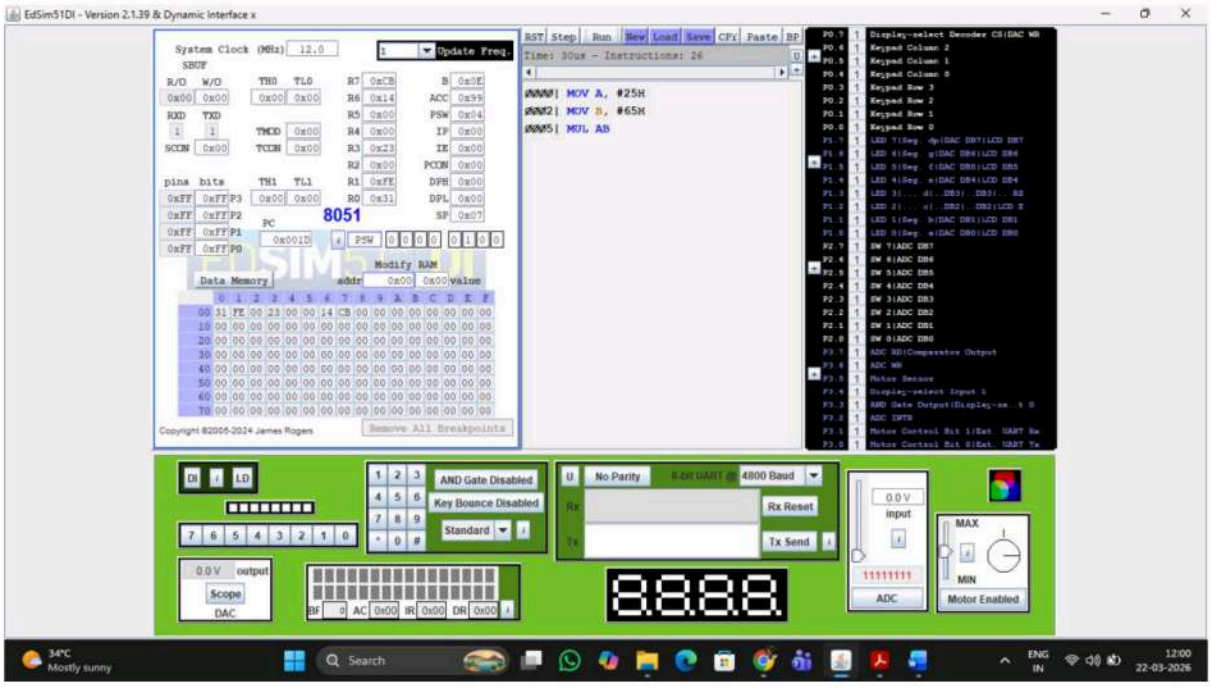
Subtraction with CY=0:



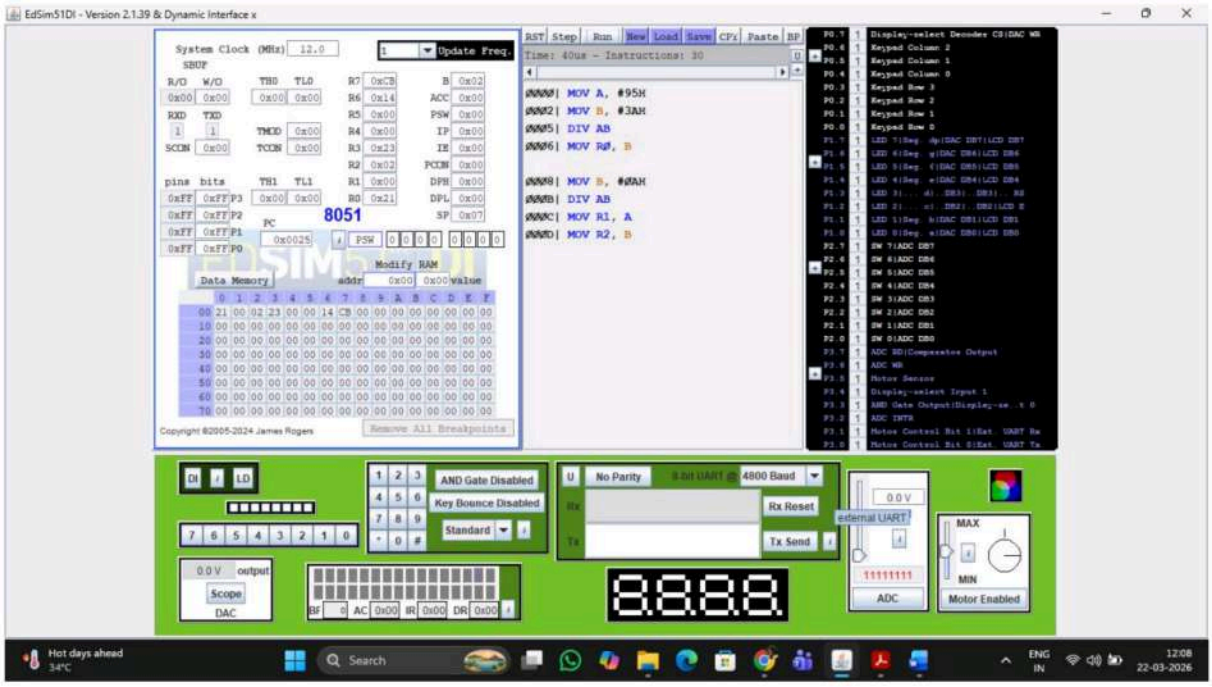
Subtraction with CY=1:



MULTIPLICATION:



DIVISION:



Result:

Thus the ALU operation using 8051 Microcontroller was performed and the result was obtained successfully.

EX NO: DATE:	**WRITE BASIC AND ARITHMETIC PROGRAM USING EMBEDDED C**

Aim:

To write basic and arithmetic program Using Embedded C.

Apparatus required:

Keil µ software

Procedure:

- The 8051 microcontroller has a set of instructions for arithmetic, logic, and data manipulation operations. that demonstrates some basic ALU operations using assembly language for the 8051 microcontroller.
- The various ALU operations, including addition, subtraction, multiplication, division, logical AND/OR/XOR, and rotation. Note that the specific instructions and registers used may vary depending on the exact 8051 microcontroller variant you are working with.
- Always refer to the datasheet or reference manual for your specific microcontroller to ensure accurate programming.

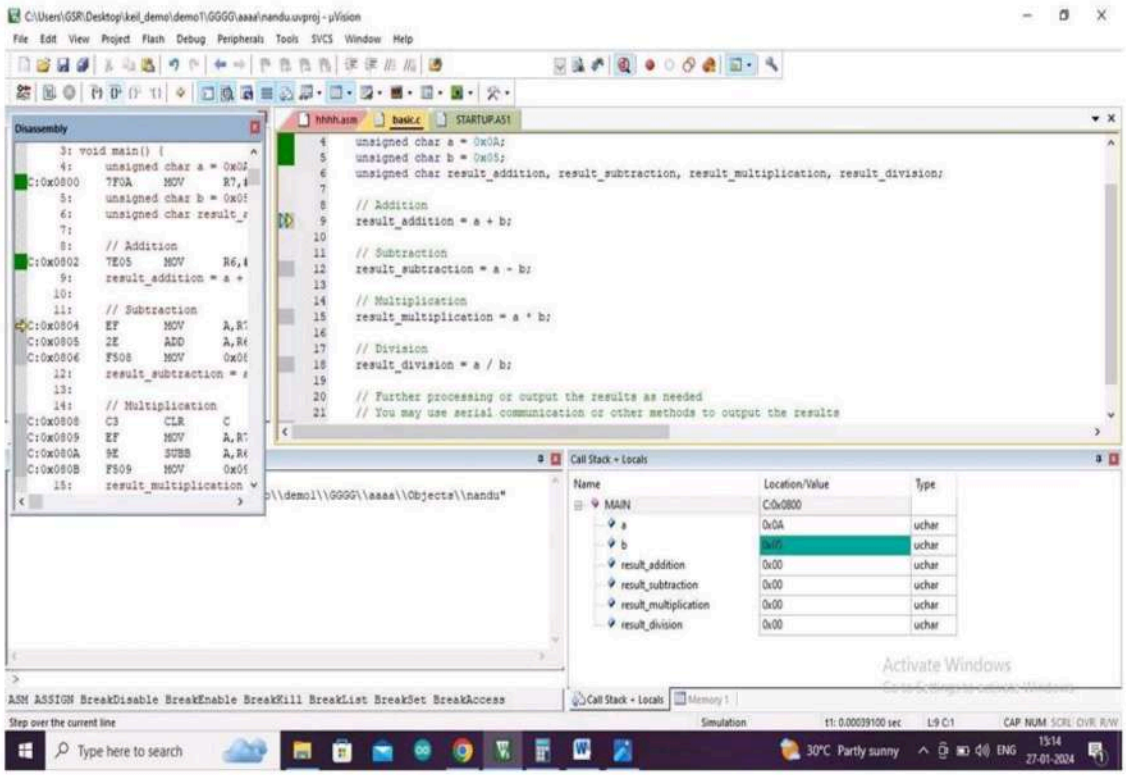
Program:

```
#include<8051.h>
void main() {
   unsigned char a=0x0A;
   unsigned char b= 0x05;
   unsigned char result_addition, result_subtraction, result_multiplication, result_division;
   result_addition = a + b;
```

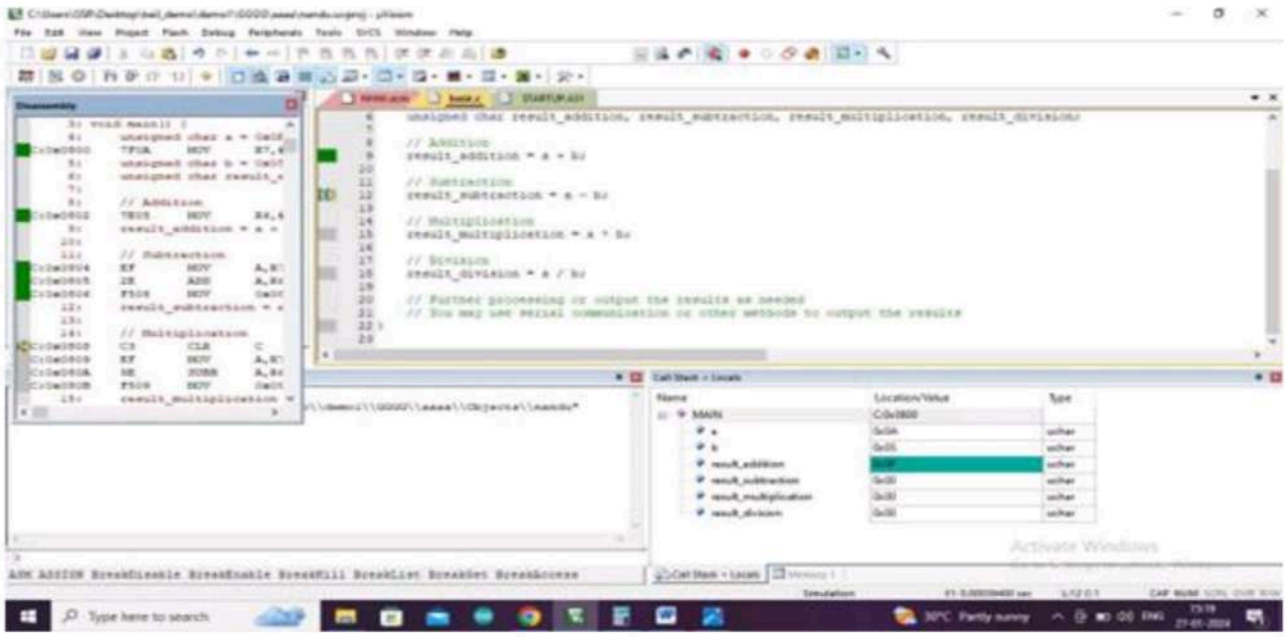
```
    result_subtraction = a –b;

    result_multiplication = a * b;

    result_division = a / b;

}
```

Output:

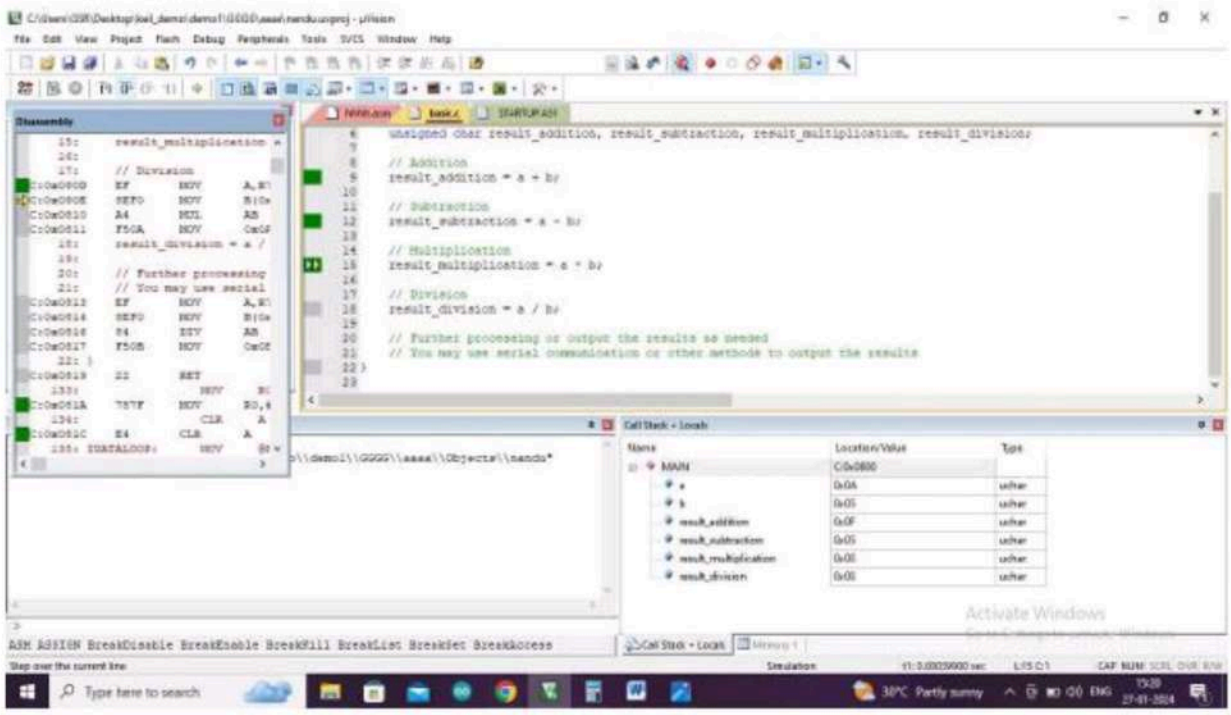
ADDITION:



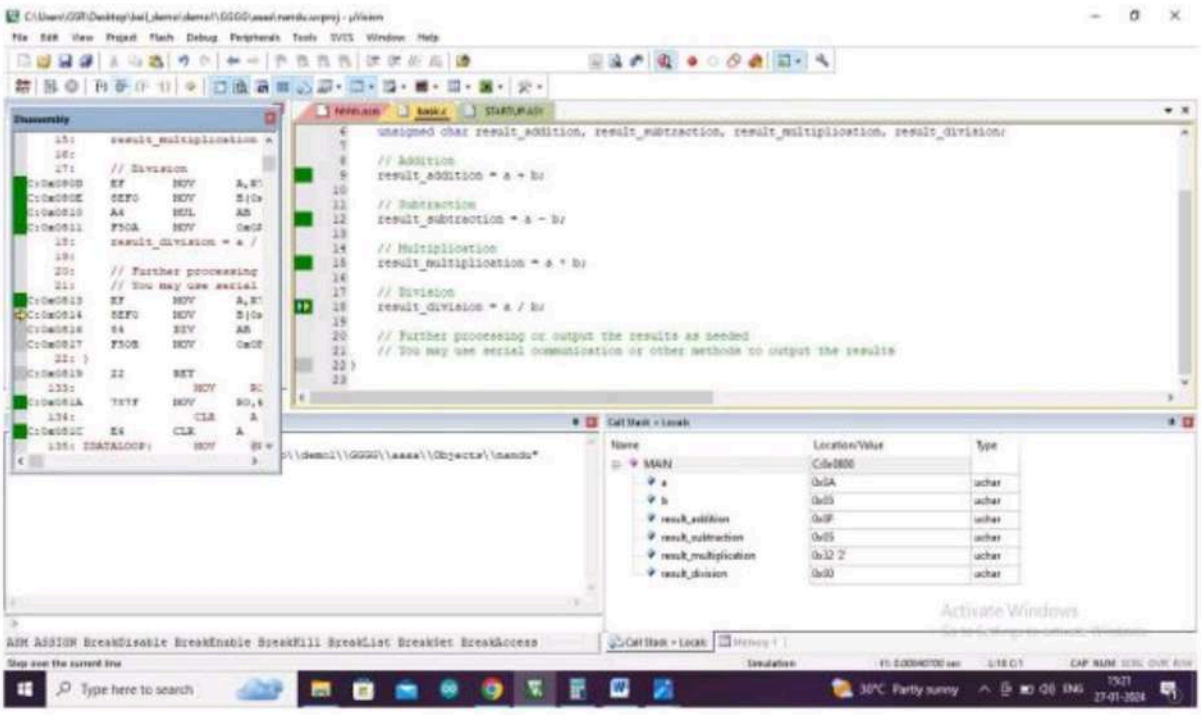
SUBTRACTION:



MULTIPLICATION:



DIVISION:



Another way

Program:

1.ADDITION:

```
#include <reg51.h>

void main(void)

{

 unsigned char x, y, z;

x = 0x12;

y = 0x34;

P0 = 0x00;

z = x + y;

P0 = z;

}
```

2.SUBTRACTION:

```
#include <reg51.h>

void main(void)

{

unsigned char a,b,c;

a=0x12;

b=0x34;

P1=0x00;

c=b-a;

P1=c;

 }
```

3.MULTIPLICATION:

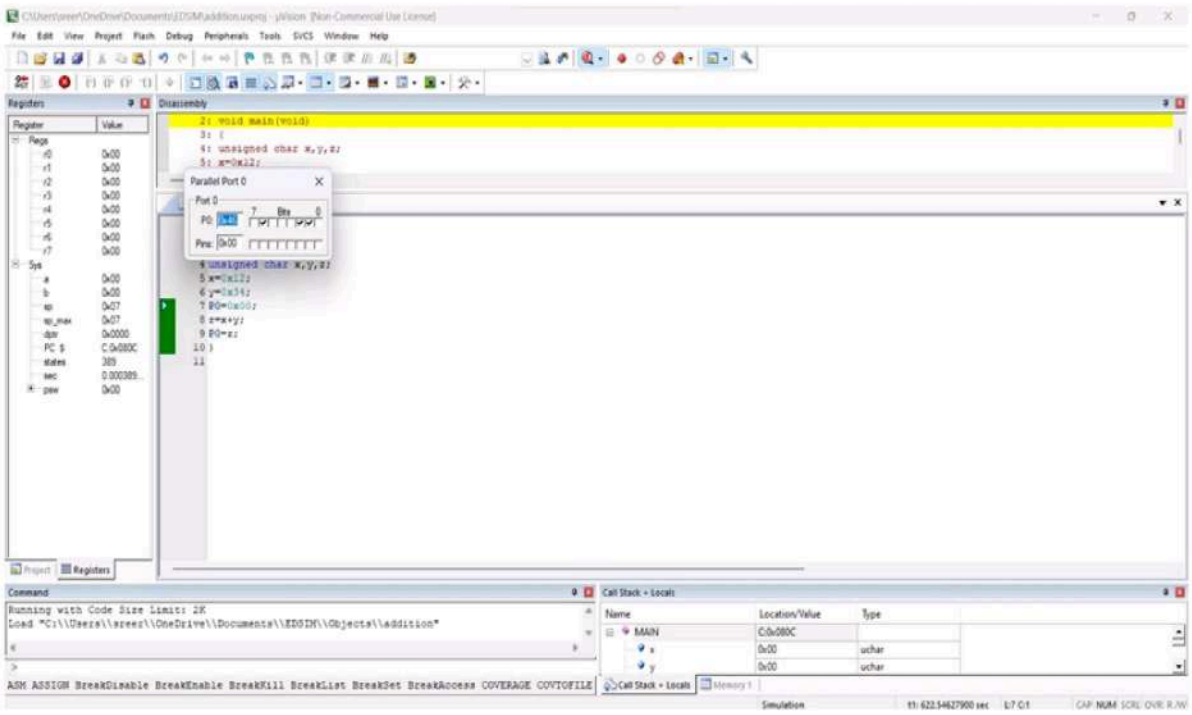
```
#include <reg51.h>
void main(void)
{
   unsigned char d, e, f;
   d = 0x12;
   e = 0x34;
   P2 = 0x00;
   f = e * d;
   P2 = f;
}
```

4.DIVISION:

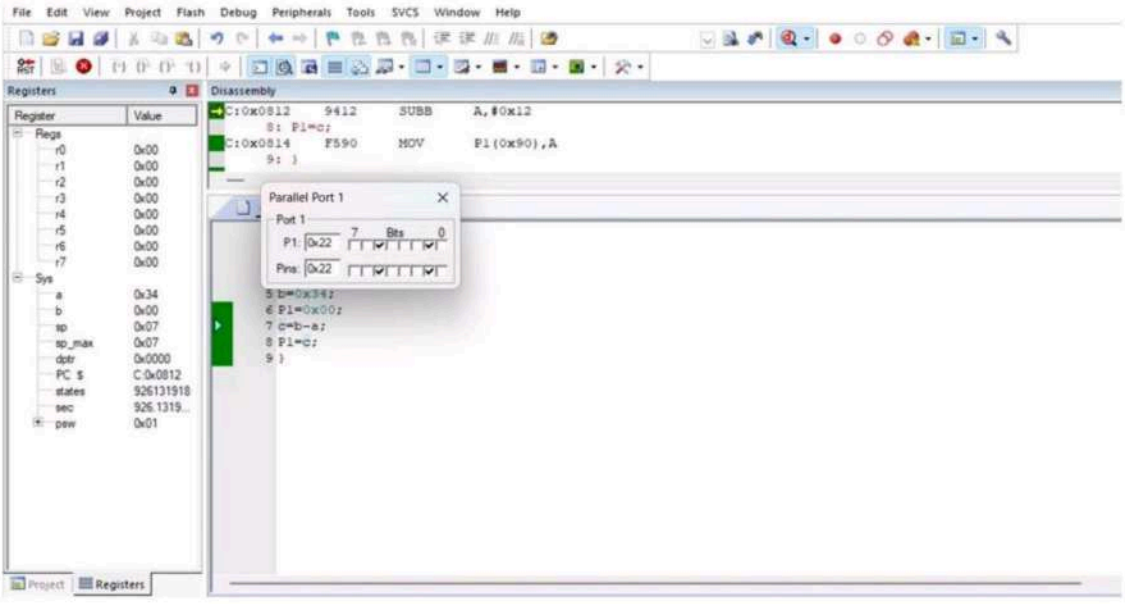
```
#include <reg51.h>
void main(void)
{
   unsigned char p, q, r;
   p = 0x12;
   q = 0x34;
   P3 = 0x00;
   r = q / p;
   P3 = r;
   while(1);
}
```

Output:

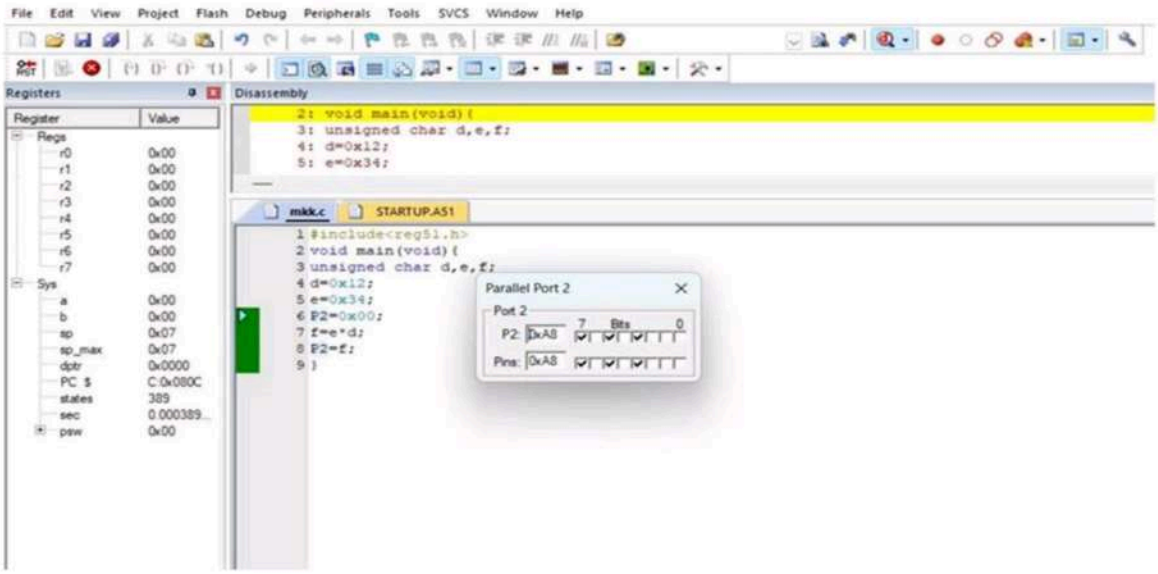
1.ADDITION:



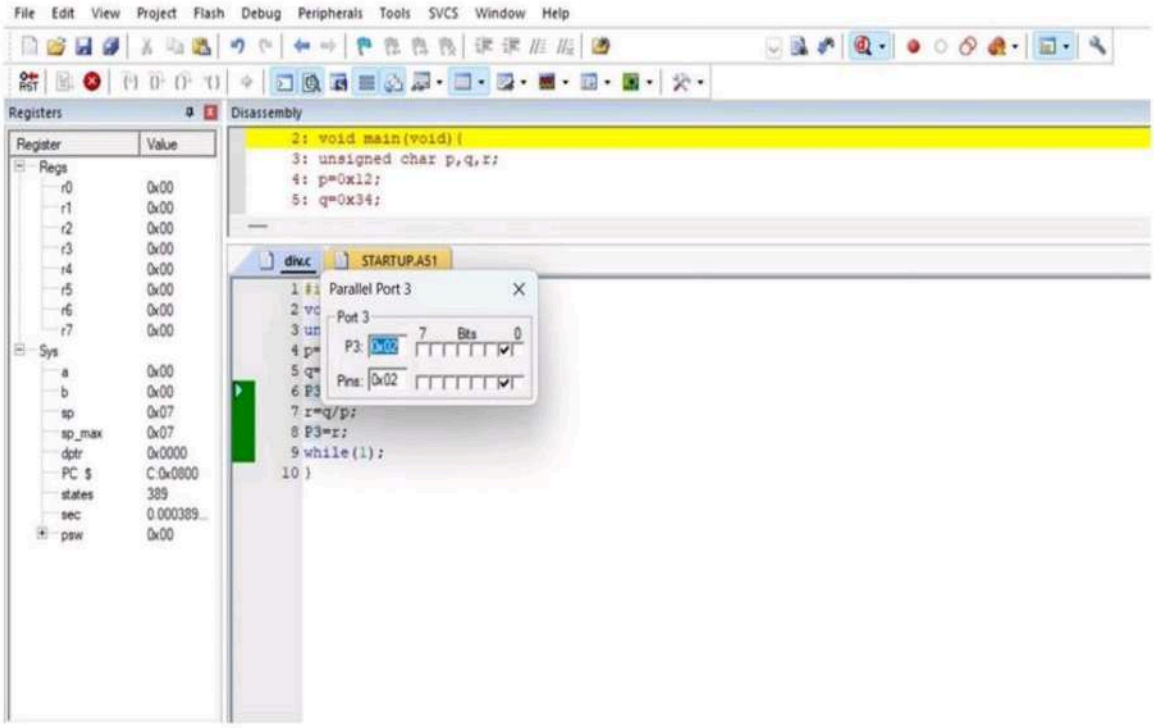
2.SUBTRACTION:



3.MULTIPLICATION:



4.DIVISION:



Result:

Thus the basic Arithmetic operations in 8051 micro-controller using Embedded- C programming was executed successfully.

EX NO : DATE :	INTRODUCTION TO ARDUINO PLATFORM AND PROGRAMMING

Aim:

To Test Arduino Ide Using Arduino Platform And Programming.

Apparatus required:

1. Arduino board (e.g., Arduino Uno)
2. Pushbutton ,RGB LED
3. Breadboard and jumper wires
4. Downloading cable

A) PUSH BUTTON USING ARDUINO

Procedure:

1. Connect your thingZkit IoT to your computer using the USB cable.
2. Connect pin 2 of the Arduino to the onboard pushbutton (SW-2) with a pull-down configuration.
3. Ensure that the connections are secure and free from any short circuits. Setup Software, Power on the thingZkit IoT board.

Program:

```
#include<Arduino.h> // Fixed missing closing '>'
#define buttonPin 2
void setup() {
 // Initialize the button pin as an input
pinMode(buttonPin, INPUT);
// Initialize serial communication
Serial.begin(9600);
Serial.println("Active-High Push Button Status:");
```

```
}
void loop() {
 // Read the status of the button bool
 buttonState = digitalRead(buttonPin);
 // Print the button status
 if (buttonState == HIGH) {
  Serial.println("Button is not pressed");
 } else {
  Serial.println("Button is pressed (Active-High)");
 }
delay(1000);
}
```

Diagrams:

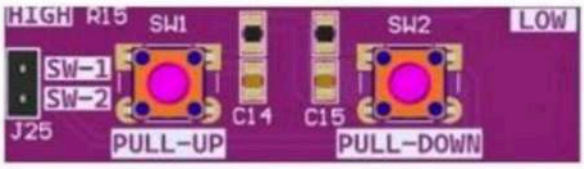
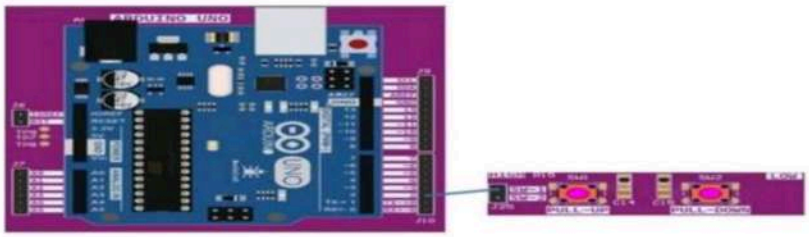


Fig no 19: General I/O Push Button

Connection Diagram:



Pin Details:

thingZkit IoT-ARDUINO	PushButton in Pull-Up Configuration
2	SW-1

Output:

B) RGB LED USING ARDUINO

Procedure:

1. Connect the RGB LED pins to Arduino digital pins 9, 10, and 11 with proper resistors.
2. Initialize the pins as OUTPUT using pinMode() in the setup function.
3. In the loop, turn ON one color at a time using digitalWrite() and turn OFF others.
4. Add delay between each color change to display red, green, blue, and white sequentially.

Program:

```
#include <Arduino.h>

/* RGB LED pins (common cathode) */

#define RED_PIN    9

#define GREEN_PIN 10

#define BLUE_PIN  11

/* delay (in milliseconds) */

#define COLOR_CHANGE_DELAY 1000

void setup() {

  pinMode(RED_PIN, OUTPUT);

  pinMode(GREEN_PIN, OUTPUT);
```

```
  pinMode(BLUE_PIN, OUTPUT);
}
void loop() {
  digitalWrite(RED_PIN, HIGH);
  digitalWrite(GREEN_PIN, LOW);
  digitalWrite(BLUE_PIN, LOW);
  delay(COLOR_CHANGE_DELAY);

  digitalWrite(RED_PIN, LOW);
  digitalWrite(GREEN_PIN, HIGH);
  digitalWrite(BLUE_PIN, LOW);
  delay(COLOR_CHANGE_DELAY);

  digitalWrite(RED_PIN, LOW);
  digitalWrite(GREEN_PIN, LOW);
  digitalWrite(BLUE_PIN, HIGH);
  delay(COLOR_CHANGE_DELAY);

  digitalWrite(RED_PIN, HIGH);
  digitalWrite(GREEN_PIN, HIGH);
  digitalWrite(BLUE_PIN, HIGH);
  delay(COLOR_CHANGE_DELAY);
  digitalWrite(RED_PIN, LOW);
  digitalWrite(GREEN_PIN, LOW);
  digitalWrite(BLUE_PIN, LOW);
  delay(COLOR_CHANGE_DELAY);
}
```

Diagram:

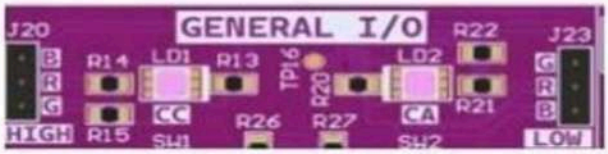
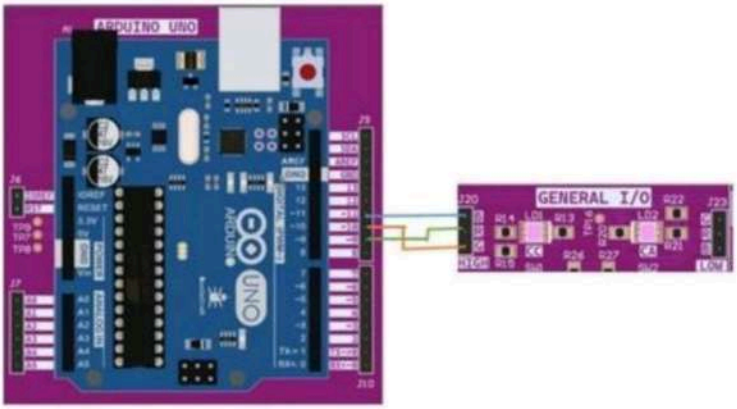


Fig no 18: General I/O RGB LED

Connection diagram:



Pin Details:

thingZkit IoT-ARDUINO	RGB LED- Common Cathode
9	Red
10	Green
11	Blue

Result:

Thus the Interfacing of button and LED with ARDUINO was performed successfully.

EX NO : DATE :	**EXPLORE DIFFERENT COMMUNICATION METHOD WITH IOT DEVICES (BLUETOOTH)**

Aim :

To write a python program to explore different communication method Bluetooth with IoT devices

Apparatus required :

arduino, arduino IDE, Bluethooth.

Procedure:

1. Connect your Arduino to your computer using the USB cable.Connect the Inbuilt Bluetooth Module with Arduino as described above Place the LED on the breadboard and connect it to the Arduino as described above.

2. Ensure that all connections are secure and free from short circuits. Setup Software: Power on the thingZkit IoT board , Open the Experiment 15 program in Arduino platform from the given source code folder.

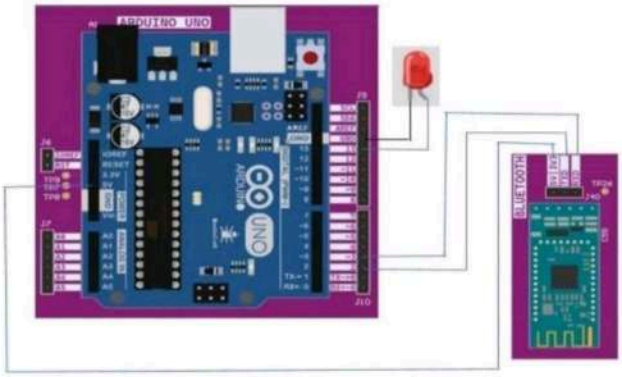
3. Install the mobile App from the Google Play Store

Program :

```
#include <SoftwareSerial.h>

#define LED_pin 13

SoftwareSerial bluetooth(2, 3); // RX (Arduino pin 2), TX (Arduino pin 3)

float data = 25.98;

char charArray[10];

void setup() {

Serial.begin(9600);        // Initialize the Serial Monitor
```

```
bluetooth.begin(9600);      // Initialize Bluetooth communication
pinMode(LED_pin, OUTPUT);   // Set LED pin as OUTPUT
}
void loop()
{    // Check if data is available from the Bluetooth module
if (bluetooth.available()) {
char receivedChar = bluetooth.read(); // Read the character received via Bluetooth
Serial.print("Received: ");
Serial.println(receivedChar);
if (receivedChar == '1')
{
digitalWrite(LED_pin, HIGH);
Serial.println("LED_ON");
}
else if (receivedChar == '0')
{
digitalWrite(LED_pin, LOW);
Serial.println("LED_OFF");
}
else if (receivedChar == '2')
{
dtostrf(data, 6, 2, charArray); // Convert float to char array
bluetooth.write(charArray);     // Send data back via Bluetooth
delay(100);
}
}
}
```

Connection:



Pin Details:

thingZkit IoT-ARDUINO	Bluetooth Module & LED
Pin 2	TX
Pin 3	RX
5V	5V/3V
Pin 13	External LED (Long leg terminal)
GND	LED Short leg terminal

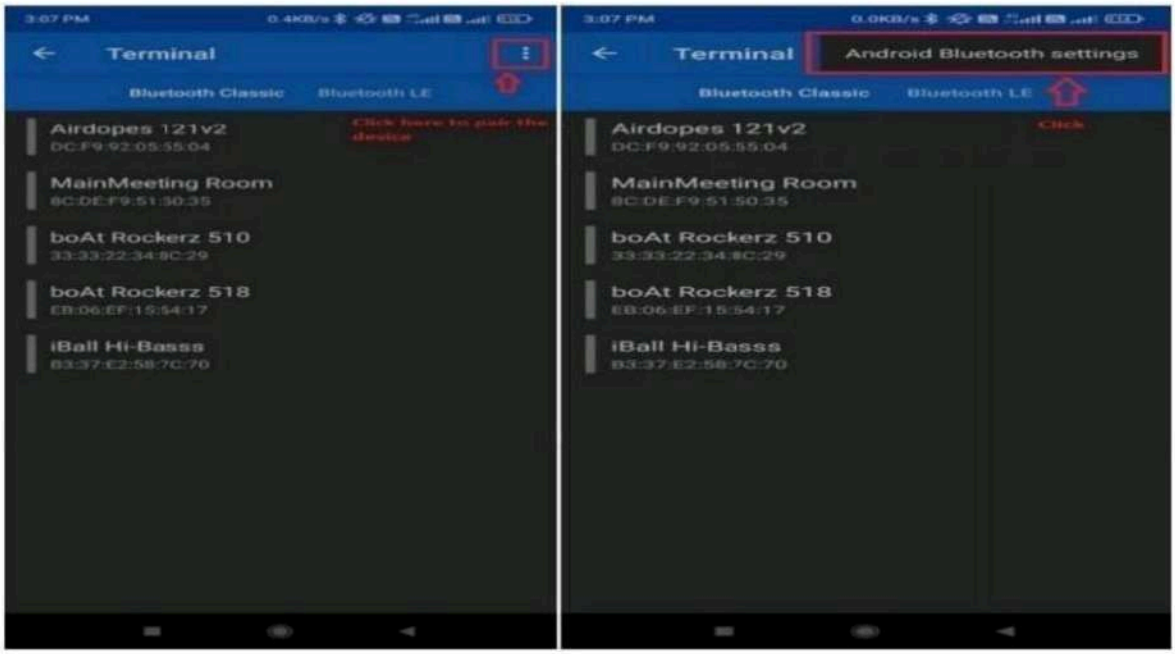
Output:

1. Open the Arduino Serial Monitor from the Arduino IDE.

2. Ensure that the baud rate in the Serial Monitor matches the baud rate in the Arduino code (in this case, 9600).

Turn on Bluetooth in mobile. Open the App and follow the steps as mentioned in the below

Screenshots:

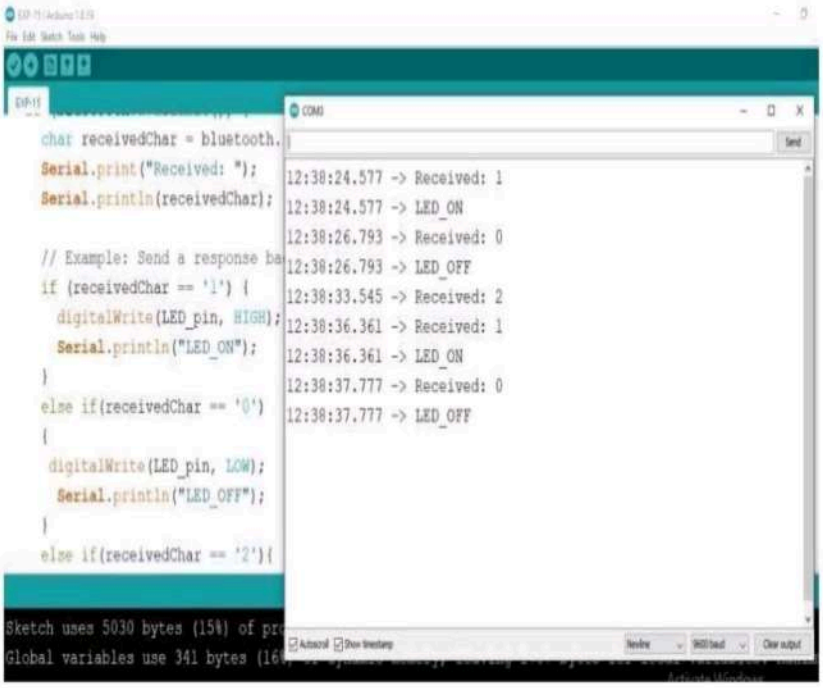

Terminal
Bluetooth Classic
Bluetooth LE
Airdopes 121v2
DC:F9:92:05:55:04
MainMeeting Room
8C:DE:F9:51:30:35
boAt Rockerz 510
33:33:22:34:8C:29
boAt Rockerz 518
EB:06:EF:15:54:17
iBall Hi-Bass
B3:37:E2:58:7C:70
Terminal
Android Bluetooth settings
Bluetooth Classic
Bluetooth LE
Airdopes 121v2
DC:F9:92:05:55:04
MainMeeting Room
8C:DE:F9:51:50:35
boAt Rockerz 510
33:33:22:34:8C:29
boAt Rockerz 518
EB:06:EF:15:54:17
iBall Hi-Basss
B3:37:E2:58:7C:70


Devices
Bluetooth Classic
Bluetooth LE
58:56:00:00:0C:F3
HC-05
HC-05
58:56:00:00:18:D3
Living Room speaker
8C:DE:F9:51:80:77
OPPO Enco M32
28:FA:19:C9:60:27
OnePlus Bullets Wire
E4:41:22:8F:95:07
OnePlus Bullets Wire
8C:64:A2:5C:71:D5
Rockerz 255 Pro
11:11:22:37:EF:EE
Vmyb
40:51:51:FC:F6:32
realme Buds Air 3
D8:6C:02:50:54:56
Now the
and connect.


2:40 PM
0.2KB/s
Terminal
14:39:50.664 Connecting to ESP32 ...
14:39:51.384 Connected
14:39:54.724 Disconnected
14:39:55.724 Connecting to ESP32 ...
14:39:55.814 Connected
Check the status as "connected"
M1
M2
M3
M4
M5
M6
M7


Terminal
12:37:19.963 Connecting to HC-05 ...
12:37:20.518 Connected
on
off
M3
M4
M5
M6


2:42 PM
Terminal
Click here to disconnect
14:41:46.396 Connecting to ESP32 ...
14:41:47.261 Connected
14:41:52.146 on
14:41:57.157 off
14:42:04.821 Disconnected
M1
M2
M3
M4
M5
M6
M7



Result:

Thus the python program to explore different communication method with IoT devices (Bluetooth) was written and executed successfully

EX NO : DATE :	INTRODUCTION TO RASPBERRY PI PLATFORM AND PYTHON PROGRAMMING

Aim:

Connect a LED with Raspberry Pi with thonny python ide of the Raspberry Pi.

Appartus required :

- Raspberry pi board
- Blinking LED
- Power supply
- Cables
- Internet Connectivity

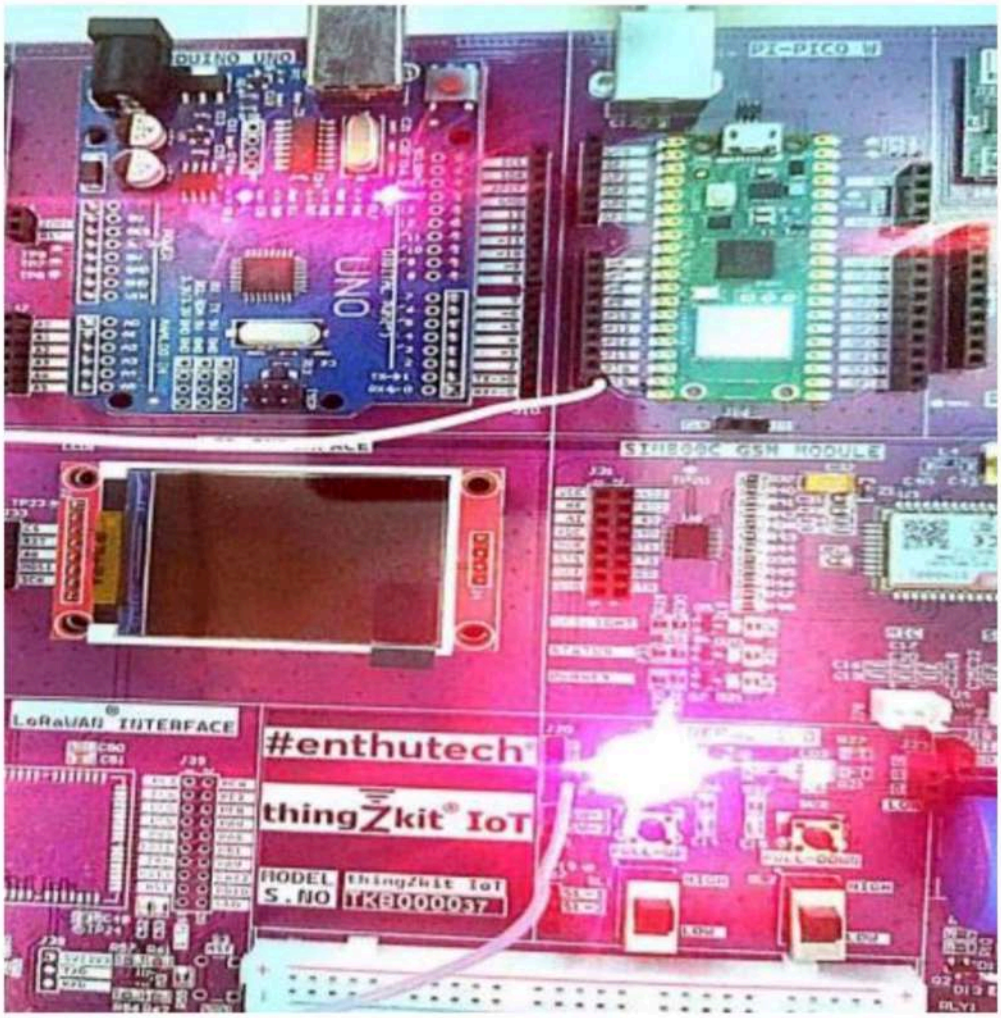
Procedure:

1. Using thonny python ide and create code in python format.
2. Connect GPIO 17 to LED.
3. Run the python ide

Program:

```
from machine import Pin
import time
led = Pin(17, Pin.OUT)  # GPIO 17 on Pico
while True:
    led.value(1)
    print("LED ON")
    time.sleep(1)
    led.value(0)
    print("LED OFF")
    time.sleep(1)
```

Output:



Result:

Thus the Interfacing of LED with Raspberry Pi platform was performed successfully.

EX NO : DATE :	INTERFACING SENSOR WITH RASPBERRY PI

Aim:

Interfacing the ultrasonic sensor with a Raspberry Pi involves connecting the sensor to the GPIO pins of the Raspberry Pi.

Apparatus required:

- Raspberry pi board
- ultrasonic sensor
- Blinking LED
- power supply
- cables
- Internet Connectivity

Procedure:

- Connect the VCC pin of the ultrasonic sensor to the 5V pin on the Raspberry Pi.
- Connect the TRIG pin of the ultrasonic sensor to a GPIO 14 pin on the Raspberry Pi.
- Connect the ECHO pin of the ultrasonic sensor to the GPIO 15 pin on the Raspberry Pi.
- Save the Python script with a .py extension .
- Run the python ide. Verify the distance.

Program:

```
from machine import Pin, time_pulse_us
import time
# Define pins
```

```
TRIG = Pin(14, Pin.OUT)  # example: GPIO14
ECHO = Pin(15, Pin.IN)   # example: GPIO15
def get_distance():
    # Make sure trigger is low
    TRIG.value(0)
    time.sleep_us(2)
    # Trigger high for 10us
    TRIG.value(1)
    time.sleep_us(10)
    TRIG.value(0)
    # Measure the echo pulse
    duration = time_pulse_us(ECHO, 1, 30000)
    if duration < 0:
        return None
    distance_cm = (duration / 2) * 0.0343
    return distance_cm
    while True:
        distance = get_distance()
        if distance is not None:
            print("Distance: {:.2f} cm".format(distance))
        else:
            print("Out of range")
        time.sleep(1)
```

Output:

```
from machine import Pin, time_pulse_us
import time

# Define pins
TRIG = Pin(14, Pin.OUT)  # example: GPIO3
ECHO = Pin(15, Pin.IN)   # example: GPIO2

def get_distance():
    # Make sure trigger is low
    TRIG.value(0)
    time.sleep_us(2)

    # Trigger high for 10us
    TRIG.value(1)
    time.sleep_us(10)
    TRIG.value(0)

    # Measure the echo pulse
    duration = time_pulse_us(ECHO, 1, 30000)  # timeout 30ms
    if duration < 0:
        return None  # Timeout occurred

    # Calculate distance (speed of sound: 343 m/s)
    distance_cm = (duration / 2) * 0.0343
    return distance_cm

# Main loop
while True:
    distance = get_distance()
    if distance is not None:
        print("Distance: {:.2f} cm".format(distance))
    else:
        print("Out of range")
    time.sleep(1)
```

```
Shell ×
Distance: 244.27 cm
Distance: 243.58 cm
Distance: 244.04 cm
Distance: 244.66 cm
Distance: 244.18 cm
Distance: 244.34 cm
```

Result:

Thus the python program to interface ultrasonic sensors with Raspberry PI was written and executed successfully.

EX NO : DATE :	**STUDY OF ARDUINO TO PI WIRELESS COMMUNICATION**

Aim :

To study wireless communication between arduino and Pi pico W .

Apparatus required :

1. Arduino UNO.
2. PICO W .
3. Jumpers
4. TFT Display.

Procedure :

1. UPLOAD TX Code to PI PICO W using Thonny python IDE .
2. Upload Rx Code to Arduino UNO using Arduino IDE.
3. Change the wifi SSID and PASSWORD in both code .
4. Ones connected enter the IP address for server in both codes.
5. Connect as per the connection diagram.
6. Run the code .

Program :

Tx Code In PICO W:

```
import network

import socket

import time
```

```
# Wi-Fi credentials
ssid = 'WIFINAME'
password = 'PASSWORD'
# Server details (Arduino with ESP8266/ESP32)
server_ip = '192.168.238.10'   # Replace with your server IP
server_port = 80               # Must match Arduino server port
# Connect to Wi-Fi
wlan = network.WLAN(network.STA_IF)
wlan.active(True)
wlan.connect(ssid, password)
# Wait until connected
while not wlan.isconnected():
    print("Connecting to Wi-Fi...")
    time.sleep(1)
print("Connected to Wi-Fi")
print(wlan.ifconfig())  # Print IP configuration
# Create TCP socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
# Connect to server
try:
    client_socket.connect((server_ip, server_port))
    print("Connected to the server.")
except Exception as e:
    print("Failed to connect to the server:", e)
    client_socket.close()
```

```
# Message to send
send_message = "tq team"
# Send data continuously
while True:
    try:
        # Send message
        client_socket.send(send_message.encode('utf-8'))
        print("Sent:", send_message)

        # Receive acknowledgment
        ack = client_socket.recv(1024)
        print("Received:", ack.decode('utf-8'))
        time.sleep(3)
    except Exception as e:
        print("Error:", e)
        client_socket.close()
        break
# Close socket
client_socket.close()
```

Rx Code for Arduino UNO:

```
#include <SoftwareSerial.h>
#define RX 2
#define TX 3
SoftwareSerial esp8266(RX, TX);
// Wi-Fi credentials
String AP   = "WIFINAME";
```

```
String PASS = "PASSWORD";


// Server configuration
String PORT = "80";
int countTrueCommand;
int countTimeCommand;
boolean found = false;
void setup() {
  Serial.begin(9600);
  esp8266.begin(115200);
  // Initialize ESP8266
  sendCommand("AT", 5, "OK");
  // Set mode: SoftAP + Station
  sendCommand("AT+CWMODE=3", 5, "OK");
  // Connect to Wi-Fi
  sendCommand("AT+CWJAP=\"" + AP + "\",\"" + PASS + "\"", 80, "OK");
  // Enable multiple connections
  sendCommand("AT+CIPMUX=1", 5, "OK");
  // Start server
  sendCommand("AT+CIPSERVER=1," + PORT, 5, "OK");
  Serial.println("Server started. Waiting for clients...");
}
void loop() {
  if (esp8266.available()) {
String response = esp8266.readString();
    Serial.println("Received data: " + response);
    // Check for incoming data
    if (response.indexOf("+IPD") >= 0) {
```

```
 int start = response.indexOf(":") + 1;
      String data = response.substring(start);
      Serial.println("Message: " + data);
    }
  }
}

// Function to send AT commands
void sendCommand(String command, int maxTime, char readReplay[]) {
  Serial.print(countTrueCommand);
  Serial.print(". AT command => ");
  Serial.print(command);
  Serial.print(" ");
  while (countTimeCommand < (maxTime * 1)) {
    esp8266.println(command);
    if (esp8266.find(readReplay)) {
      found = true;
      break;
    }
    countTimeCommand++;
  }
  if (found) {
    Serial.println("OK");
    countTrueCommand++;
    countTimeCommand = 0;
  } else {
    Serial.println("Failed");
```

```
    countTrueCommand = 0;
      countTimeCommand = 0;


    }
    found = false;
  }
```

Result :

Thus Tested was Successfully Completed.

EX NO : DATE :	**STUDY OF SETUP A CLOUD PLATFORM TO LOG THE DATA**

Aim:

To study to setup a cloud platform to log the data.

Apparatus Required:

1. ARDUINO IDE
2. NODE MCU
3. IR SENSOR
4. Bread board

Procedure:

Step 1: Connect the IR sensor to the microcontroller.

Step 2: Create a channel in ThingSpeak and get the API key.

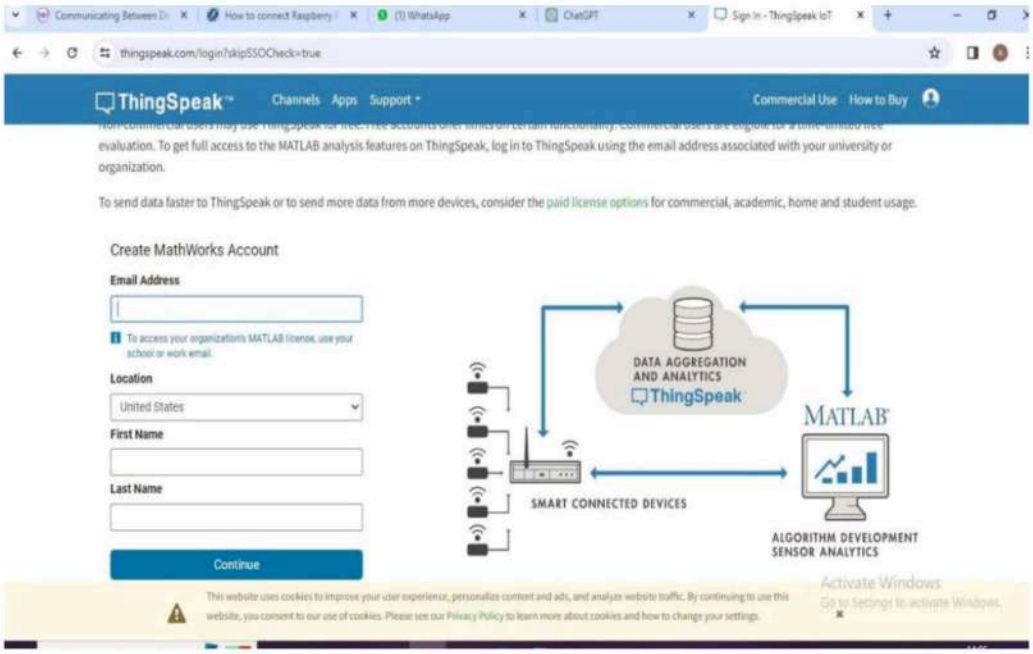
Step 3: Install Arduino IDE and required libraries.

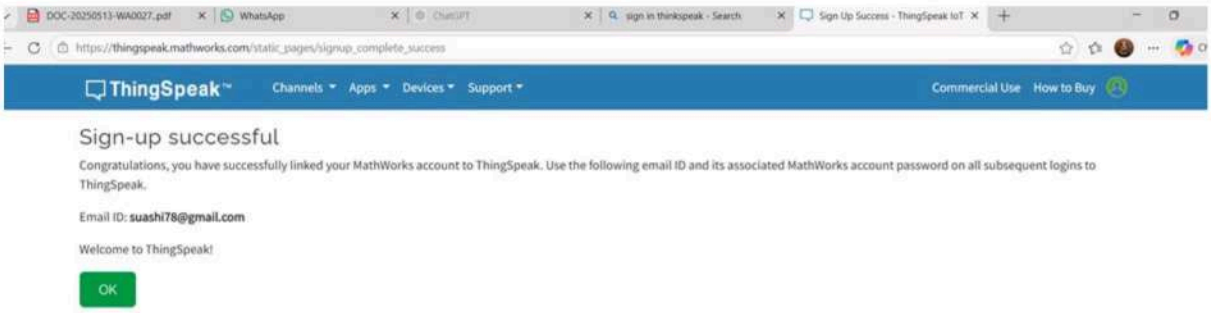
Step 4: Write code to read IR sensor data and connect to Wi-Fi.

Step 5: Send data to ThingSpeak and upload the code.

Step 6: View the data on the ThingSpeak graph

Screenshots:



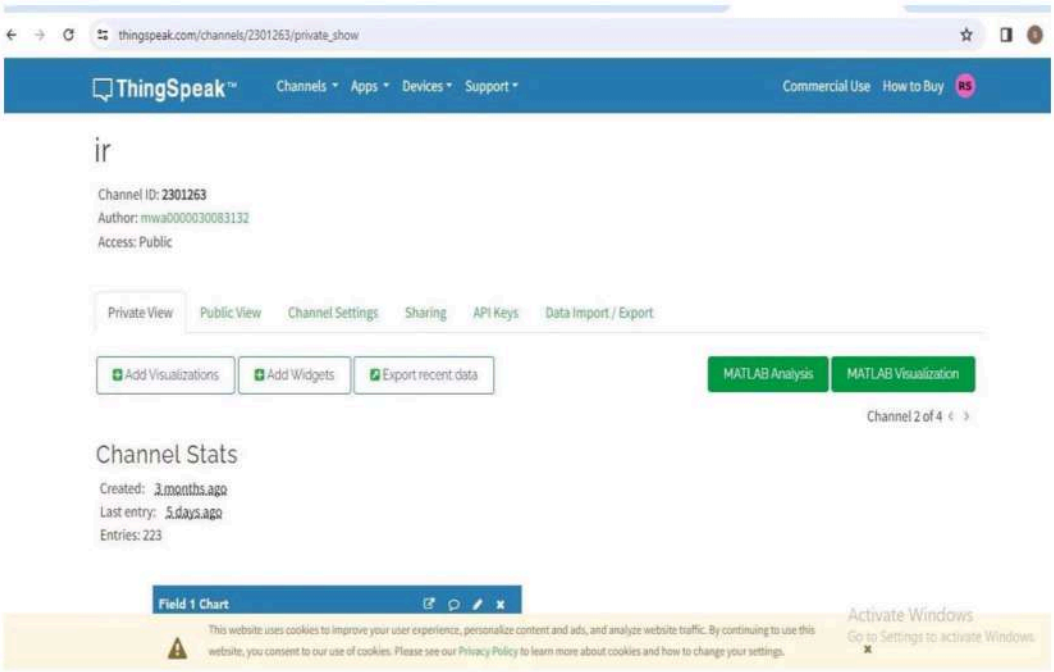


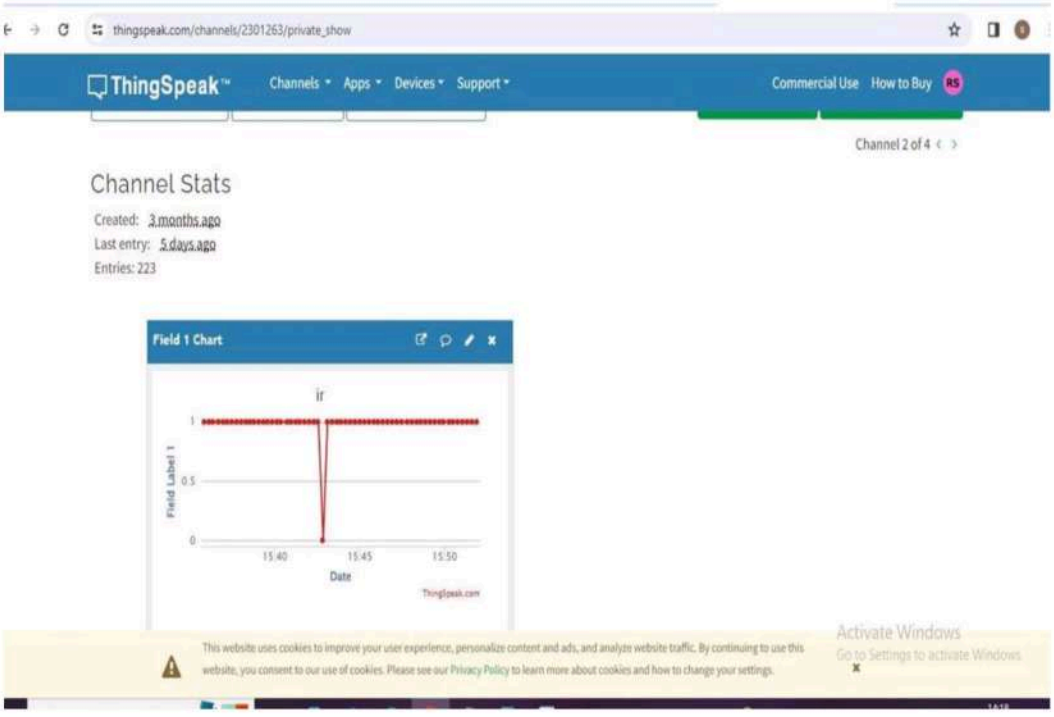
Program :

```
#include <IRremote.h>

#include <ESP8266WiFi.h>

#include <ThingSpeak.h>

const char *ssid = "silicon systems";

const char *password = "Silicon@2017";

const char *apiKey = "QWD5TTQ08RBGQFVH";
```

```
const int IR_PIN = 2; // Change this to the pin connected to your IR sensor

IRrecv irrecv(IR_PIN); decode_results results;

void setup() {   Serial.begin(9600);

irrecv.enableIRIn(); // Start the IR receiver

WiFi.begin(ssid,password);

ThingSpeak.begin(client); // Initialize ThingSpeak

}

void loop() {

if (irrecv.decode(&results)) {

// Read and send the data to ThingSpeak

int irValue = results.value;

Serial.print("IR Value: "); Serial.println(irValue);

// Send data to ThingSpeak

ThingSpeak.writeField(YOUR_CHANNEL_ID, 1, irValue, apiKey);

irrecv.resume(); // Receive the next value

delay(10000); // Delay to avoid sending data too frequently

}

// Other loop logic can be added here

}
```

Output:





Result :

The IR sensor data was successfully sent to ThingSpeak and displayed in graphical form for real-time monitoring.

EX NO : DATE :	**STUDY OF LOG DATA USING RASPERRY PI AND UPLOAD TO THE CLOUD PLATFORM**

Aim :

To study test the log data using raspberry pi and upload to the cloud platform.

Apparatus required:

1. RASPBERRY PI
2. NODE MCU
3. IR SENSOR
4. Bread board

Procedure :

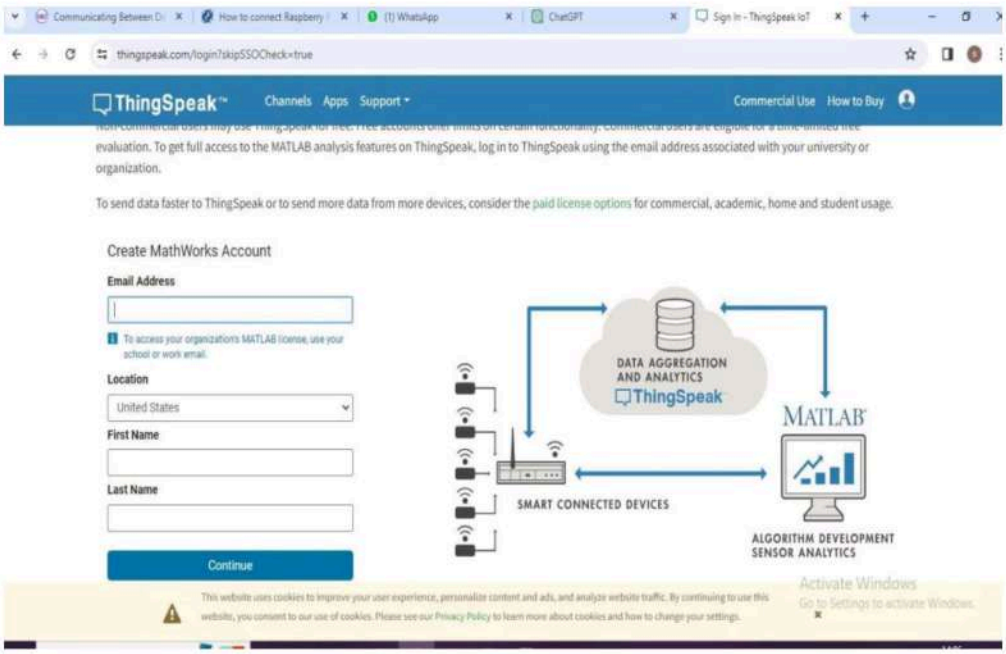
1. Connect the IR sensor to the Arduino.
2. The connections may vary based on your specific IR sensor model, but typically, it involves connecting the sensor's signal pin to a digital pin on the Arduino and power/ground pins appropriately.
3. Install the necessary libraries for the IR sensor.
4. For example, if you are using a common IR sensor like the TSOP382, you might need the "IRremote" library.

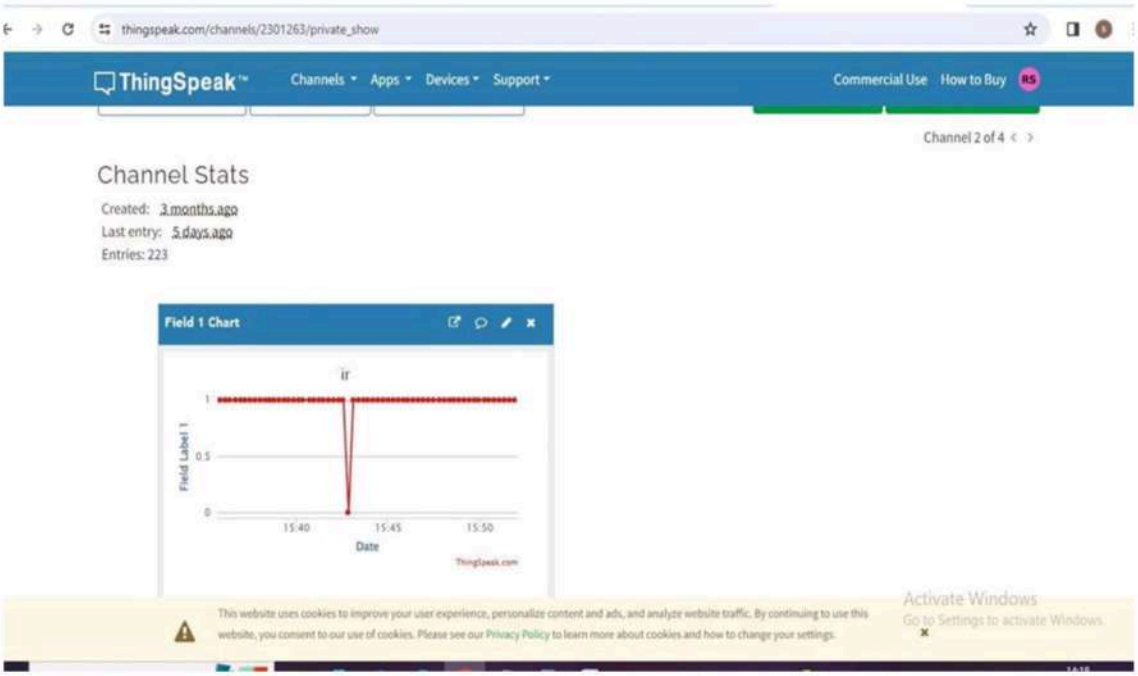
Program :

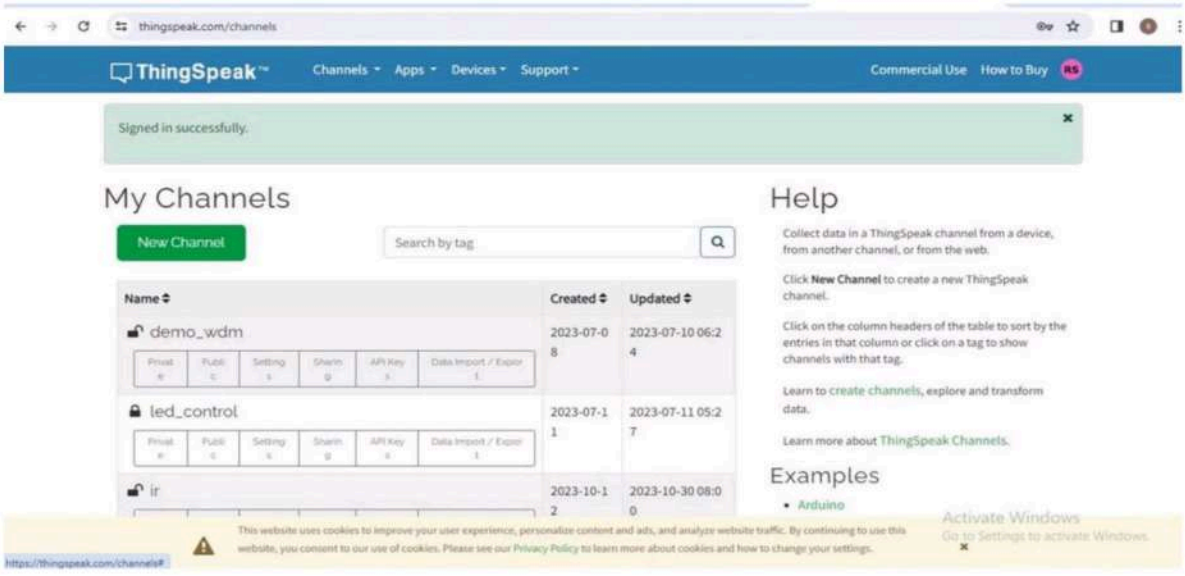
```
import machine

import utime

import urequests

import network

IR_SENSOR_PIN = 2 # GPIO pin for the IR sensor

API_KEY = "ZSX5SRTSRJ61XVU6"

THINGSPEAK_URL =

"https://api.thingspeak.com/update?api_key={}".format(API_KEY)

WIFI_SSID = "@ms.Balaji"

WIFI_PASSWORD = "@msbalaji"

ir_sensor = machine.Pin(IR_SENSOR_PIN, machine.Pin.IN)

wlan = network.WLAN(network.STA_IF)

def connect_to_wifi(): wlan.active(True)

if not wlan.isconnected(): print("Connecting to WiFi...")

wlan.connect(WIFI_SSID, WIFI_PASSWORD) while not wlan.isconnected():

pass

print("Connected to WiFi")

def read_ir_sensor():

    return ir_sensor.value()

def send_to_thingspeak(data):

# Send data to ThingSpeak

url = "{}&field1={}".format(THINGSPEAK_URL, data)

response = urequests.get(url)

print("ThingSpeak response:", response.text)

# Connect to WiFi connect_to_wifi()
```

```
while True:
    ir_data = read_ir_sensor()
    print("IR Sensor Data:", ir_data)
# Send data to ThingSpeak via HTTP
send_to_thingspeak(ir_data)
utime.sleep(15)  # Wait for 15 seconds before the next reading
```

Output :







Result :

Thus the study of log data using Raspberry Pi and upload to the cloud platform was Successfully tested and Completed.